Identification and Stability of SDF Estimators for Pooled, Non-traded Cash Flows

Christian Tausch
AssetMetrix GmbH
Theresienhöhe 13, D-80339 Munich
christian.tausch@quant-unit.com
Tel: +49895432880333
*Corresponding Author

Alexander Bohnert
Munich University of Applied Sciences
Am Stadtpark 20, D-81243 Munich
alexander.bohnert@hm.edu

Abstract

We develop a semiparametric Least-Mean-Distance framework for identifying stochastic discount factors from pooled, non-traded cash flows under cross-sectional dependence induced by overlapping payoff streams. The framework nests the inception-based estimator of Driessen et al. (2012) in a setting with increasing-domain asymptotics. We show that averaging pricing errors over additional discounting dates beyond the initial investment stabilizes finite-sample estimation but shifts the population target through a timing-related covariance mechanism. Applied to private-equity cash-flow panels, the estimator yields stable market-beta estimates but only weakly identifies abnormal performance or multifactor exposures.

Keywords: Stochastic Discount Factor, Semiparametric Estimation, Spatial Dependence, Generalized Method of Moments (GMM), Private Equity, Illiquid Assets

JEL classification: G12, G24, C14, C58

Introduction

How should stochastic discount factor (SDF) estimators be identified and interpreted when the underlying data consist of pooled, non-traded cash flows rather than traded returns? This question arises in a growing private-market asset-pricing literature concerned with abnormal performance and systematic risk in settings such as private equity (see Korteweg, 2019, 2023, for comprehensive reviews). The semiparametric SDF estimators of Driessen et al. (2012) and Korteweg and Nagel (2016), now standard tools in that literature, recover abnormal performance (α) and factor exposures (β) from pooled cash flows without strong distributional assumptions. But an important practical issue remains unresolved: which economically meaningful objects can those estimators identify with stability in the sparse, overlapping, and partially NAV-based panels that researchers actually observe?

This paper studies that question by formalizing a tradeoff that is implicit in SDF estimation with pooled, non-traded cash flows. In typical private-capital fund samples, inception-only pricing

errors, as in Driessen et al. (2012), can be highly unstable because the objective function is not globally convex in finite samples and the effective cross-sectional sample size is limited. A natural response is to average pricing errors over additional compounding dates beyond inception. But this is not merely a statistical refinement: once pricing errors are evaluated at horizons beyond inception, the estimator generally targets a different population object. We formalize this mechanism as the *Compounding-Horizon Wedge* and show that horizon choice creates a tradeoff between finite-sample stabilization and population target shift. At the same time, the simulation evidence below shows that this estimand shift need not be purely a cost: over an intermediate range of horizons, horizon averaging can improve finite-sample recoverability not only by lowering dispersion, but also by reducing higher-order small-sample bias of the estimator around its horizon-specific target.

We develop this framework in the context of private-equity cash flows and apply it to Preqin data on 894 Buyout and 931 Venture Capital funds (North America, USD, vintages 1982–2010). The empirical results deliver three main findings. First, parsimonious single-factor market models yield comparatively stable estimates across a broad range of compounding horizons, with market betas of roughly 0.8–1.2 for Buyout and 1.0–1.9 for Venture Capital, depending on horizon and weighting scheme. Second, once an α intercept or a second factor is introduced, estimates become highly unstable: market betas oscillate, sign reversals occur, and asymptotic standard errors become erratic, consistent with weak identification rather than a well-determined multifactor structure. Third, including unseasoned post-2010 vintages with large reported NAV components mechanically inflates estimated market betas (for Buyout, from about 1.2 to 2.4), underscoring that sample construction is itself an identification choice in private-equity asset pricing.

These findings do not imply that private-equity alpha is zero or that multifactor models are wrong; they indicate that semiparametric alpha estimates from currently available pooled cash-flow panels are weakly identified and highly sensitive to implementation choices. That qualification matters for interpreting both positive and negative findings on private-equity outperformance across the broader literature.

The paper makes three contributions.

First, we characterize identification in semiparametric cash-flow-based SDF estimation under flexible horizon choice. We derive a generalized pooled-cash-flow pricing framework that makes explicit how the choice of discounting date affects the estimand, nesting the inception-only criterion of Driessen et al. (2012) as a special case. This formulation reveals a distinction that, to our knowledge, has not been formalized: a compounding horizon may improve finite-sample performance without preserving the original inception-based pricing restriction. The framework also clarifies the relation to Korteweg and Nagel (2016). Their estimator is closely connected but not a literal special case because the order of aggregation differs, so that the two approaches generally converge to different pseudo-true parameter vectors under misspecification.

Second, we formulate the estimator as a nonlinear Least-Mean-Distance (LMD) problem for pooled, non-traded cash flows (Pötscher and Prucha, 1997) and develop an increasing-domain asymptotic framework for inference under dependence induced by overlapping funds and adjacent vintages. This fills a gap in the existing literature: Driessen et al. (2012) use within-portfolio infill asymptotics that do not generate new aggregate-state realizations, while Korteweg and Nagel (2016) rely on nonoverlapping payoff windows that are scarce in typical panels. Our vintage-year increasing-domain device is better aligned with unconditional identification when new vintages bring new macroeconomic realizations. In the empirical application, however, we treat finite-sample simulation evidence and stability diagnostics as the primary basis for interpretation whenever asymptotic approximations appear unreliable. In that sense, this is an identification and stability paper for private-market asset pricing.

Third, we provide simulation and empirical evidence on the practical boundaries of semiparametric SDF identification for private equity. We validate the estimator's finite-sample behavior in Monte Carlo experiments calibrated to private-equity-like cash-flow environments and implement hv-block cross-validation diagnostics (Racine, 2000) tailored to overlapping cash-flow dependence. The empirical application demonstrates that, in our sample, current cash-flow panels support stable inference about broad market exposure in seasoned vintage-year portfolios but not about abnormal performance or richer factor decompositions.

Related literature. The paper contributes to the semiparametric private-equity performance literature that builds on Driessen et al. (2012) and Korteweg and Nagel (2016). The Driessen et al. (2012) estimator has been widely applied to price human capital contracts, venture capital, real estate, and private equity funds (Kroencke et al., 2013; Kim et al., 2015; Farrelly and Stevenson, 2019; Lee et al., 2019; Zhao, 2020; Tausch and Pietz, 2024). The Korteweg and Nagel (2016) framework has found broad application spanning private equity, infrastructure, REITs, CLOs, impact funds, and private debt (Giommetti and Jørgensen, 2021; Gupta and Nieuwerburgh, 2021; Andonov et al., 2021; Riddiough and Wiley, 2022; Boyer et al., 2023; Cordell et al., 2023; Erel et al., 2024; Jeffers et al., 2024; Korteweg and Nagel, 2024; Korteweg et al., 2024). Our LMD framework contributes by providing a loss-function formulation that encompasses Driessen et al. (2012) and contextualizes Korteweg and Nagel (2016) while supplying rigorous increasing-domain asymptotics (drawing on Pötscher and Prucha (1997), Kim and Sun (2011), and Jenish and Prucha (2012)). While Gredil et al. (2020) highlighted that NPV-based measures are biased relative to per-period returns, we explicitly formalize this discrepancy as the Compounding-Horizon Wedge, a concept related to discount-rate risk in secondary markets (Boyer et al., 2023).

More broadly, the paper connects to parametric private-equity models (Franzoni et al., 2012; Ang et al., 2018), the NAV-smoothing literature (Brown et al., 2019; Jenkinson et al., 2020; Brown et al., 2023), credit-market approaches to private-firm valuation (Hüther et al., 2022), and the SDF performance evaluation tradition in public equities (Farnsworth et al., 2002) and hedge funds (Li et al., 2016). From an econometric perspective, the paper is also complementary to Burnside (2016), who studies identification and inference in linear SDF models with traded excess returns under alternative normalizations. In our pooled non-traded cash-flow setting, the key issue is instead that horizon averaging regularizes the sample criterion while shifting the population pricing restriction.

The remainder of the paper proceeds as follows. Section 1 develops the generalized semi-parametric estimator and the corresponding asymptotic inference framework. Section 2 discusses implementation choices and studies finite-sample behavior in simulation. Section 3 presents the

empirical application to Buyout and Venture Capital funds. Section 4 concludes.

1 Methodology

We now formalize a semiparametric framework for SDF estimation with pooled cash flows based on the seminal approach of Driessen et al. (2012). Section 1.1 establishes the theoretical asset-pricing conditions and highlights how the choice of discounting date induces the Compounding-Horizon Wedge. Section 1.2 uses this insight to introduce a generalized Least-Mean-Distance estimator that endogenizes the discounting dates. Section 1.3 summarizes the increasing-domain asymptotic benchmark and SHAC inference used to interpret the empirical results under overlapping vintage-year dependence. Section 1.4 demonstrates how this LMD estimator nests the framework of Driessen et al. (2012) as a special case and relates to Korteweg and Nagel (2016).

1.1 Asset pricing for pooled cash flows

Because private equity funds invest capital over multiple quarters, any fund-level pricing restriction must aggregate deal-level moments across different investment dates. This subsection asks under what conditions such a pooled cash-flow stream satisfies a valid stochastic-discount-factor pricing restriction. The central finding is that the choice of discounting date is not innocuous: discounting at or before the first investment date preserves a clean conditional zero-moment condition (Lemma 1, Part 1), whereas post-investment discounting or future-value compounding generally introduces a timing-related covariance wedge (Parts 2 and 3). The argument proceeds in three steps. First, we state a deal-level decomposition of fund cash flows (Assumption 1) and derive the timing result at the deal level (Lemma 1). Second, we aggregate to the fund level and show that pooled Net Present Value (NPV) moments satisfy a valid zero conditional moment restriction whenever discounting occurs at or before fund inception (Proposition 1). Third, we discuss the implications for Net Future Value (NFV) moments and the interpretation of alpha measured ex post. Throughout, the deal-level structure serves only as a latent accounting device used to illuminate and justify the fund-level

restriction; only pooled fund cash flows are required as observable inputs to the estimator.

For fund i, let $CF_{i,t}$ denote the observed net cash flow at time t, and define the fund inception date by $t_i^0 := \min_j t_{i,j}^{\text{Inv}}$. All fund-level variables are constructed from the following deal-level data generation process.

Assumption 1. ***Deal-level decomposition.*** *For each fund i, there exist J_i underlying deals such that:*

1. *Deal j has exactly one negative investment cash flow $\text{Inv}_{i,j} < 0$ at time $t_{i,j}^{\text{Inv}} \in \{0, 1, \ldots, T-1\}$.*
2. *Deal j has a finite stream of positive divestment cash flows $(\text{Div}_{i,j,k})_{k=1,\ldots,K_{i,j}}$ with $\text{Div}_{i,j,k} > 0$ and $t_{i,j,k}^{\text{Div}} > t_{i,j}^{\text{Inv}}$ for all k.*
3. *Let $cf_{i,j,t}$ denote the deal-level cash flow of deal j in fund i at time t. Then the observed fund-level cash flow is the cross-deal aggregate*

$$CF_{i,t} = \sum_{j=1}^{J_i} cf_{i,j,t} \qquad \forall i, t.$$

4. *The fund has no cash flows before inception: $CF_{i,t} = 0$ for all $t < t_i^0$.*

Under our central asset-pricing restriction, each deal is correctly priced by the stochastic discount factor at its investment date. Formally, the deal-level pricing error satisfies

$$\mathbb{E}\left[\delta_{i,j} \middle| \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right] = 0 \quad \forall\, i, j \tag{1}$$

where the deal-level pricing error is defined by

$$\delta_{i,j} := \text{Inv}_{i,j} + \sum_{k=1}^{K_{i,j}} \frac{\Psi_{t_{i,j,k}^{\text{Div}}}}{\Psi_{t_{i,j}^{\text{Inv}}}} \text{Div}_{i,j,k} \tag{2}$$

and the multi-period SDF $\Psi_t = \prod_{u=1}^{t} M_u$ discounts a time-t cash flow to time 0 (and $\Psi_0 = 1$).

The next result characterizes how this deal-level pricing restriction depends on the choice of discounting or compounding date.

Lemma 1. *Suppose Equation* (1) *holds.*

1. *If* $\tau \leq t_{i,j}^{\text{Inv}}$*, then*

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\delta_{i,j}\,\middle|\,\mathcal{F}_\tau\right] = 0. \tag{3}$$

2. *If* $\tau > t_{i,j}^{\text{Inv}}$*, then*

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\delta_{i,j}\,\middle|\,\mathcal{F}_\tau\right] = \frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\,\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_\tau\right], \tag{4}$$

   which need not equal zero.

3. *Under the same pre-investment timing as Part (1), suppose additionally that the pricing error is compounded forward to a date* $t_{i,j}^{\text{Future}} > t_{i,j}^{\text{Inv}}$*. If* $\tau \leq t_{i,j}^{\text{Inv}} < t_{i,j}^{\text{Future}}$*, then define the NFV-type moment*

$$m_{i,j}^{\text{NFV}}(\tau, t_{i,j}^{\text{Future}}) := \frac{\Psi_{t_{i,j}^{\text{Future}}}}{\Psi_\tau}\,\delta_{i,j}.$$

   Its expected value

$$\mathbb{E}\left[m_{i,j}^{\text{NFV}}(\tau, t_{i,j}^{\text{Future}}) \mid \mathcal{F}_\tau\right] = \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\text{Cov}\left(\frac{\Psi_{t_{i,j}^{\text{Future}}}}{\Psi_{t_{i,j}^{\text{Inv}}}}, \delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right) \mid \mathcal{F}_\tau\right] \tag{5}$$

   generally contains a conditional covariance term and therefore need not equal zero.

Proof. Consider part (1). Applying the law of iterated expectations, since $\tau \leq t_{i,j}^{\text{Inv}}$, the SDF ratio $\Psi_{t_{i,j}^{\text{Inv}}}/\Psi_\tau$ is $\mathcal{F}_{t_{i,j}^{\text{Inv}}}$-measurable and can be factored out, yielding

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\delta_{i,j} \mid \mathcal{F}_\tau\right] = \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\underbrace{\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right]}_{=0 \text{ by Equation (1)}} \mid \mathcal{F}_\tau\right] = 0.$$

For part (2), since $t_{i,j}^{\text{Inv}} < \tau$, the ratio $\Psi_{t_{i,j}^{\text{Inv}}}/\Psi_\tau$ is $\mathcal{F}_\tau$-measurable, so

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\delta_{i,j}\,\middle|\,\mathcal{F}_\tau\right] = \frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_\tau\right].$$

Since some divestment cash flows of deal j may have been paid by date τ, $\delta_{i,j}$ is partially $\mathcal{F}_\tau$-measurable, and its conditional expectation given $\mathcal{F}_\tau$ generally reflects realized outcomes rather than the zero-mean restriction at $t_{i,j}^{\text{Inv}}$.

For part (3), using the law of iterated expectations with an intermediate conditioning on $\mathcal{F}_{t_{i,j}^{\text{Inv}}}$,

$$\begin{aligned}\mathbb{E}\left[m_{i,j}^{\text{NFV}}(\tau, t_{i,j}^{\text{Future}}) \mid \mathcal{F}_\tau\right] &= \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Future}}}}{\Psi_{t_{i,j}^{\text{Inv}}}}\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right] \mid \mathcal{F}_\tau\right] \\ &= \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\text{Cov}\left(\frac{\Psi_{t_{i,j}^{\text{Future}}}}{\Psi_{t_{i,j}^{\text{Inv}}}}, \delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right) \mid \mathcal{F}_\tau\right]\end{aligned}$$

where we used the conditional covariance identity and Equation (1). □

Part (3) of Lemma 1 shows that compounding deal cash flows to a future date generally destroys the zero conditional moment restriction because the resulting expression contains a covariance between the pricing error and future SDF realizations. We refer to this non-zero expected value as a *timing-related covariance wedge*. Economically, it is the time-τ price of an investment-date covariance term between continuation SDF realizations and the discounted divestment cash flows over the illiquid holding period. At the level of the generalized estimator, this mechanism induces the **Compounding-Horizon Wedge**: once the criterion averages over post-investment or future-value revaluation dates, the timing-related covariance wedge enters the pooled objective and shifts the population target away from the inception-based benchmark.

To clarify the economic mechanism behind the Compounding-Horizon Wedge, let us briefly analyze Equation (5) with only one divestment cash flow per deal ($K_{i,j} = 1$). In this case, the

covariance term reads

$$\mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Future}}}}{\Psi_{t_{i,j}^{\mathrm{Inv}}}},\ \mathrm{Inv}_{i,j} + \frac{\Psi_{t_{i,j,k}^{\mathrm{Div}}}}{\Psi_{t_{i,j}^{\mathrm{Inv}}}}\mathrm{Div}_{i,j,k} \mid \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\right) = \mathrm{Cov}\left(\prod_{u=t_{i,j}^{\mathrm{Inv}}+1}^{t_{i,j}^{\mathrm{Future}}} M_u,\ \mathrm{Div}_{i,j,k} \prod_{u=t_{i,j}^{\mathrm{Inv}}+1}^{t_{i,j,k}^{\mathrm{Div}}} M_u \mid \mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\right)$$

Thus, for $K_{i,j} = 1$, the covariance is naturally expected to be largest when $t_{i,j}^{\mathrm{Future}}$ is equal to $t_{i,j,k}^{\mathrm{Div}}$, since the continuation SDF and the discounted divestment payoff then load on the same SDF path. This ranking is not automatic, however, and requires additional restrictions on the post-divestment SDF dynamics (e.g, $\Psi_{t_{i,j}^{\mathrm{Future}}}/\Psi_{t_{i,j}^{\mathrm{Div}}}$ being conditionally mean-independent of $\mathcal{F}_{t_{i,j}^{\mathrm{Div}}}$ given $\mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}$).

To conclude our analysis of Part 3 of Lemma 1, NFV-type moments are inappropriate as primitive ex-ante pricing moments unless the timing-related covariance wedge is modeled explicitly. They can still be informative for ex-post benchmarking, but the resulting performance measure must then be interpreted as a composite object that mixes true abnormal performance (skill) with systematic exposure to the states in which realization and discounting occur.

Proposition 1. ***Price of a pooled cash-flow stream at or before fund inception.*** *If $\tau \leq t_i^0$, then*

$$\mathbb{E}\left[\sum_{t=\tau}^{T} \frac{\Psi_t}{\Psi_\tau} CF_{i,t} \,\middle|\, \mathcal{F}_\tau\right] = 0. \tag{6}$$

Proof. If $\tau \leq t_i^0$, then $\tau \leq t_{i,j}^{\mathrm{Inv}}$ for every deal j in fund i. Using the deal-level decomposition of the pooled cash-flow stream and Equation (2),

$$\mathbb{E}\left[\sum_{t=\tau}^{T} \frac{\Psi_t}{\Psi_\tau} CF_{i,t} \,\middle|\, \mathcal{F}_\tau\right] = \mathbb{E}\left[\sum_{j=1}^{J_i} \frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau} \delta_{i,j} \,\middle|\, \mathcal{F}_\tau\right]. \tag{7}$$

Applying Equation (3) term-by-term yields

$$\mathbb{E}\left[\sum_{j=1}^{J_i} \frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau} \delta_{i,j} \,\middle|\, \mathcal{F}_\tau\right] = 0. \tag{8}$$

□

Proposition 1 provides the clean benchmark for identification: pooled NPV moments are theoretically well founded when discounting occurs at or before fund inception. Once $\tau > t_i^0$, some underlying deals are active or already partially realized, so the corresponding pooled pricing error incorporates the Compounding-Horizon Wedge. The choice of discounting dates is therefore not innocuous. Section 1.2 formalizes this choice by defining an estimator that operates entirely on pooled fund cash flows (the deal-level decomposition having served its purpose as an analytical device) and explicitly specifies which discounting dates are used. When it restricts to dates $\tau \leq t_i^0$, the pooled fund-level pricing error inherits a valid zero conditional moment. When it includes dates $\tau > t_i^0$, it generally targets a tilted criterion that embeds the Compounding-Horizon Wedge characterized by Equation (5).

Critically, the Compounding-Horizon Wedge is not solely relevant for ex-post benchmarking. It also fundamentally drives the large-sample and small-sample properties of our generalized LMD estimator. As we demonstrate in later sections, averaging pricing errors over extended horizons deliberately introduces wedge-bearing discounting dates that smooth the nonlinear sample criterion and can materially improve finite-sample behavior, even though doing so shifts the population target.

1.2 Least-Mean-Distance estimator

In this subsection, we define a generalized fund-level criterion that averages pricing errors across a user-specified set of discounting dates $\mathcal{T}_i$. This nests the inception-only NPV criterion of Driessen et al. (2012) as a special case and allows us to study how horizon averaging trades structural validity against finite-sample regularization. Throughout, only pooled fund cash flows are observed, whereas the deal-level decomposition from Section 1.1 remains a latent theoretical device.

The stochastic discount factor $\Psi_{\tau,t} = \Psi_t/\Psi_\tau$ is used to calculate the time-τ price $P_{\tau,t,i}$ of a **single** time-t cash flow of any given PE fund i

$$P_{\tau,t,i} := \Psi_{\tau,t} \cdot CF_{i,t} = \frac{\Psi_t}{\Psi_\tau} \cdot CF_{i,t} \qquad \forall \quad \tau, t, i \tag{9}$$

with again multi-period SDF $\Psi_t = \prod_{u=1}^{t} M_u$ and $\Psi_0 = 1$. As SDFs are commonly parameterized by a vector $\theta \in \mathbb{R}^p$, i.e., $\Psi_{\tau,t} \equiv \Psi_{\tau,t}(\theta)$, our goal is to find an estimation method for the true parameter vector θ_0.

For each fund i and all points τ within a common fund lifetime, the empirical pricing error $\epsilon_{\tau,i}(\theta)$ of **all** realized-to-date fund cash flows is calculated as a time-τ revaluation of the entire cash-flow stream

$$\epsilon_{\tau,i}(\theta) := \sum_{t=0}^{T} P_{\tau,t,i}(\theta) \qquad \forall \quad \tau, i \tag{10}$$

Note that for $t < \tau$, the terms $\frac{\Psi_t}{\Psi_\tau} CF_{i,t}$ are compounded past cash flows, not discounted future cash flows, so the term "present value" is no longer accurate for $\tau > t_i^0$. For $\tau > t_i^0$, $\epsilon_{\tau,i}(\theta)$ is not a continuation value but a time-τ revaluation of the entire realized cash-flow stream, and therefore generally inherits the Compounding-Horizon Wedge characterized in Section 1.1. This time-τ revaluation $\epsilon_{\tau,i}(\theta)$ must be sharply distinguished from the latent continuation value $LCV_{i,\tau}(\theta) := \mathbb{E}[\sum_{t=\tau}^{T}(\Psi_t/\Psi_\tau)CF_{i,t} \mid \mathcal{F}_\tau]$, which conditions on information at τ and is not directly observed.

To analyze the effect of different discounting dates τ on the bias and variance of the estimated SDF parameters, we define the generalized criterion $\bar{\epsilon}_i(\theta)$ that averages pricing errors over the set $\mathcal{T}_i$:

$$\bar{\epsilon}_i(\theta) = w_i \cdot \frac{1}{\text{card}(\mathcal{T}_i)} \sum_{\tau \in \mathcal{T}_i} \epsilon_{\tau,i}(\theta) \qquad \forall \quad i \tag{11}$$

where w_i normalizes fund-level pricing errors before the loss is applied; it is not an observation weight in the outer sample average. Depending on the application, w_i may normalize by invested capital, by aggregate vintage-year capital, by a macroeconomic deflator, or be set to one when cash flows are already expressed in comparable scale units.

The role of $\mathcal{T}_i$ The set $\mathcal{T}_i$ controls which discounting dates enter the averaged pricing error $\bar{\epsilon}_i(\theta)$ in Equation (11) (see Figure 1). Averaging over multiple dates in $\mathcal{T}_i$ was originally motivated by the "exploding alpha" problem described by Driessen et al. (2012): when the initial cash flow is very small, an arbitrarily large alpha can discount all subsequent cash flows close to zero, rendering

the beta factors irrelevant and distorting estimation. Thus, their inception-only objective function is not globally convex in finite samples: the multiplicative SDF parameterization permits extreme parameter regions in which the discount factors drive all NPV_i close to zero, creating spurious near-minima that can prevent reliable identification of α. Averaging over $\mathcal{T}_i$ mitigates this degeneracy (see Figure 2). The relevant horizon family is parameterized as

$$\mathcal{T}_{i,H} = \left\{t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max})\right\}, \qquad H \geq 0. \tag{12}$$

This formulation encompasses two economically distinct cases:

- $H = 0$: If $\mathcal{T}_i = \{t_i^0\}$, the estimator uses an inception-only NPV criterion, is anchored in the pooled pricing restriction from Section 1.1, and targets the structural parameter θ_0.
- $H > 0$: If $\mathcal{T}_i = \mathcal{T}_{i,H}$ with $H > 0$, the estimator is a generalized horizon-averaged criterion and targets the pseudo-true parameter $\theta_H^\dagger$. It acts as a finite-sample regularization device by smoothing the sample criterion, but it does so by introducing the Compounding-Horizon Wedge and therefore changing the estimand.

Larger values of the horizon H offer more regularization, but a larger departure from the structural benchmark with zero expected value.

For $\mathcal{T}_{i,H} = \{t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max})\}$, the estimator targets the horizon-specific pseudo-true parameter $\theta_H^\dagger$, which coincides with the structural parameter θ_0 only when $H = 0$. For $H > 0$, the estimator trades structural fidelity against finite-sample regularization by averaging pricing errors over discounting dates that may incorporate the Compounding-Horizon Wedge from Section 1.1. Supplementary Appendix B analyzes this target shift mechanism in rigorous detail, providing an exact Mean Squared Error (MSE) decomposition that mathematically formalizes how $\mathcal{T}$-averaging dampens finite-sample criterion volatility at the cost of shifting the population pseudo-true target point.

Restriction used for identification and asymptotic theory. In the consistency analysis below, we specialize the generalized criterion to $\mathcal{T}_i = \{t_i^0\}$. Under Proposition 1, this yields a valid pooled fund-level zero-moment restriction at the structural parameter θ_0. This restriction guarantees that $\mathbb{E}[\bar{\epsilon}_i(\theta_0)] = 0$ under the true parameter. For the quadratic loss $L(x) = x^2$, the population objective $Q_0(\theta) = -\mathbb{E}[\bar{\epsilon}_i(\theta)^2] = -\mathrm{Var}[\bar{\epsilon}_i(\theta)] - (\mathbb{E}[\bar{\epsilon}_i(\theta)])^2$ involves both variance and bias terms, so consistency requires that θ_0 uniquely maximizes this composite criterion (see Proposition 2).

LMD estimator. Because our estimator is defined through the optimization of a sample criterion, it is an extremum estimator in the sense of Newey and McFadden (1994, Eq. 1.1), with $\hat{\theta} = \arg\max_{\theta\in\Theta} Q_n(\theta)$. Concretely, our Least Mean Distance (LMD) estimator (Pötscher and Prucha, 1997, Equation 7.1) minimizes the average loss of $\bar{\epsilon}_i(\theta)$. For consistency with the extremum-estimator convention, we will technically maximize the average negative loss of $\bar{\epsilon}_i(\theta)$, which is equivalent.

$$\hat{\theta} = \arg\min_{\theta\in\Theta} \ \frac{1}{n}\sum_{i=1}^{n} L\big(\bar{\epsilon}_i(\theta)\big) \equiv \arg\max_{\theta\in\Theta} \ Q_n(\theta) \tag{13}$$

with

$$Q_n(\theta) = -\frac{1}{n}\sum_{i=1}^{n} L\big(\bar{\epsilon}_i(\theta)\big)$$

where $L : \mathbb{R}^1 \to \mathbb{R}^1$ denotes a loss function, e.g., $L(x) = (x-0)^2$ in the case of nonlinear least squares. Throughout the paper, the weighted average fund pricing error $\bar{\epsilon}_i \equiv \bar{\epsilon}_i(\theta)$ is regarded as a nonlinear random function of the SDF parameter θ.

1.3 Asymptotic benchmark and SHAC inference

The asymptotic theory plays a supporting role in the paper. It provides a large-sample benchmark for the horizon-indexed criterion actually being estimated. It also clarifies the population object associated with each horizon and motivates the dependence-robust covariance estimator used later in the empirical analysis. Because the available private-equity panels remain small and

heavily overlapping, we do not treat asymptotic approximations as the primary source of empirical credibility in finite samples.

The large-sample analysis is stated for the horizon-indexed estimator

$$\hat{\theta}_H = \arg\max_{\theta\in\Theta} Q_{n,H}(\theta), \qquad Q_{n,H}(\theta) := -\frac{1}{n}\sum_{i=1}^{n} L\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big), \tag{14}$$

where $\bar{\epsilon}_i^{(H)}(\theta)$ denotes the horizon-averaged pricing error based on $\mathcal{T}_{i,H}$ from Section 1.2. Throughout this subsection, the horizon parameter $H \geq 0$ is fixed while $n \to \infty$. The notation makes explicit that both the estimator and its probability limit generally depend on H.

Define the horizon-specific population objective by

$$Q_H(\theta) := \mathbb{E}\big[Q_{n,H}(\theta)\big] = -\mathbb{E}\Big[L\Big(\bar{\epsilon}_i^{(H)}(\theta)\Big)\Big], \tag{15}$$

and let the corresponding population target be

$$\theta_H^{\dagger} := \arg\max_{\theta\in\Theta} Q_H(\theta). \tag{16}$$

The asymptotic benchmark is intentionally formulated around $\theta_H^{\dagger}$, not around θ_0. For $H = 0$, the pooled inception-only criterion coincides with the structural pricing restriction of Section 1.1; the benchmark target is structural when the inception-only population criterion is uniquely maximized at θ_0. For $H > 0$, the criterion generally targets a pseudo-true parameter $\theta_H^{\dagger} \neq \theta_0$, because averaging over $\tau > t_i^0$ may embed the Compounding-Horizon Wedge at the population level.

We use increasing-domain asymptotics indexed by vintage years, with local overlap dependence and bounded within-vintage clustering; the full regularity conditions are stated in Supplementary Appendix C. Under those conditions, the estimator fits the semiparametric extremum framework of Pötscher and Prucha (1997) and Newey and McFadden (1994), while the required uniform law of large numbers and random-field central limit theorem follow from Jenish and Prucha (2009, 2012).

Proposition 2 (Consistency around the horizon-specific target)**.** *Under the regularity conditions*

stated in Supplementary Appendix C,

$$\hat{\theta}_H \xrightarrow{p} \theta_H^\dagger \qquad \text{as } n \to \infty.$$

This is the horizon-indexed analogue of the standard extremum-estimator consistency result of Newey and McFadden (1994). For $H = 0$, the probability limit coincides with the structural parameter θ_0 when the inception-only population criterion is uniquely maximized at θ_0; for $H > 0$, consistency is around the horizon-specific pseudo-target $\theta_H^\dagger$.

Proposition 3 (Asymptotic normality around the horizon-specific target)**.** *Under the regularity conditions stated in Supplementary Appendix C,*

$$\sqrt{n}\left(\hat{\theta}_H - \theta_H^\dagger\right) \xrightarrow{d} \mathcal{N}(0, \Sigma_H), \qquad \Sigma_H = \mathcal{H}_H^{-1}\Lambda_H\mathcal{H}_H^{-1}.$$

This combines the extremum-estimator expansion of Newey and McFadden (1994) with the increasing-domain random-field CLT of Jenish and Prucha (2012). As with consistency, the special case $H = 0$ is centered at the structural parameter θ_0 only under the additional structural-identification condition that the inception-only population criterion is uniquely maximized at θ_0; for $H > 0$ the asymptotic center is the pseudo-target $\theta_H^\dagger$.

For large-sample inference, let $s_i^{(H)}(\theta) := \nabla_\theta L(\bar{\epsilon}_i^{(H)}(\theta))$ denote the score contribution and $\hat{\mathcal{H}}_H := \nabla_{\theta\theta} Q_{n,H}(\hat{\theta}_H)$ the sample Hessian. We estimate the long-run score covariance with the spatial HAC estimator of Kim and Sun (2011) adapted to the vintage-year metric:

$$\hat{\Lambda}_H = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} k_{i,j}\, s_i^{(H)}(\hat{\theta}_H) s_j^{(H)}(\hat{\theta}_H)^\top, \qquad k_{i,j} = K\left(\frac{d_{i,j}}{b_V}\right), \tag{17}$$

where $d_{i,j}$ is the vintage-year distance metric and b_V is the SHAC bandwidth; the full assumptions and implementation details are given in Supplementary Appendix C. The resulting asymptotic covariance matrix is

$$\widehat{\text{Avar}}(\hat{\theta}_H) = \frac{1}{n}\hat{\mathcal{H}}_H^{-1}\hat{\Lambda}_H\hat{\mathcal{H}}_H^{-1}. \tag{18}$$

Asymptotic standard errors then follow in the usual way $\mathrm{SE}(\hat{\theta}_H) = \sqrt{\mathrm{diag}\left[\widehat{\mathrm{Avar}}(\hat{\theta}_H)\right]}$.

When $H > 0$, inference based on Equation (18) pertains to the pseudo-true parameter $\theta_H^\dagger$; it therefore serves as a large-sample benchmark for stability diagnostics and model comparison, not as a direct structural test about θ_0.

Even when finite-sample inference is weak, this asymptotic benchmark still matters for interpretation. It defines the population object associated with each horizon-indexed criterion, makes the distinction between the structural parameter θ_0 and the horizon-specific pseudo-target $\theta_H^\dagger$ precise, and motivates the SHAC covariance estimator as the appropriate large-sample reference under overlap dependence. In the empirical sections, we use this benchmark to interpret, rather than override, the simulation and cross-validation evidence.

1.4 Comparison to similar estimators

Our Least-Mean-Distance (LMD) estimator belongs to the broad class of semiparametric nonlinear extremum estimators in the sense of Pötscher and Prucha (1997). Its main advantage in the present setting is that it accommodates pooled fund- or portfolio-level pricing errors, flexible horizon averaging through $\mathcal{T}_{i,H}$, and increasing-domain asymptotics indexed by vintage years, all within a comparatively transparent objective function.

This section compares it to three nearby contributions, each along a different margin. Relative to Driessen et al. (2012), our criterion nests the inception-only objective exactly, but relies on a different asymptotic device. Relative to Driessen et al. (2009), our horizon averaging addresses the same stabilization problem as ratio or log-ratio transformations, but changes the bias channel differently. Relative to Korteweg and Nagel (2016), we share the discounted-cash-flow pricing logic, but differ in aggregation and in the asymptotic source of identifying variation.

1.4.1 Connection to Driessen et al. (2012)

The connection to Driessen et al. (2012) is the closest at the objective-function level. Under the inception-only restriction, equal normalization, and quadratic loss, the LMD criterion coincides

with the identity-weighted one-step cross-sectional GMM objective used in their framework. Recall that Driessen et al. (2012) estimate $(\hat{\alpha}, \hat{\beta}) = \arg\min_{\alpha,\beta} \sum_{i=1}^{n} [NPV_i(\alpha, \beta)]^2$, where NPV_i discounts all cash flows of portfolio i to fund inception. To bridge the gap, let us rewrite NPV_i in our more general SDF notation as $PV_i(\theta) = \sum_{t \in \{t_i^0, t_i^0+1, \ldots\}} \Psi_{t_i^0,t}(\theta) CF_{i,t}$, where t_i^0 denotes the fund inception date as defined in Section 1.1.

Proposition 4 (Connection to the Driessen et al. (2012) objective)**.** *Suppose that:*

1. $\mathcal{T}_i = \{t_i^0\}$ *for all i,*
2. $w_i = 1$ *for all i,*
3. $L(x) = x^2$*,*
4. *unit i is a vintage-year portfolio.*

Then the LMD estimator solves

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} (PV_i(\theta))^2 ,$$

which is equivalent to the identity-weighted cross-sectional GMM objective in Driessen et al. (2012).

Proof. Under $\mathcal{T}_i = \{t_i^0\}$, the averaged pricing error reduces to the inception-based pooled net present value

$$\bar{\epsilon}_i(\theta) = \epsilon_{t_i^0,i}(\theta) = \sum_{t=0}^{T} \frac{\Psi_t(\theta)}{\Psi_{t_i^0}(\theta)} CF_{i,t} = PV_i(\theta).$$

With $w_i = 1$ and $L(x) = x^2$, Equation (13) becomes

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} (PV_i(\theta))^2 .$$

Writing the vector of pooled pricing errors as $e(\theta) = (PV_1(\theta), \ldots, PV_n(\theta))^\top$, the objective is proportional to $e(\theta)^\top e(\theta)$, which is the identity-weighted one-step GMM criterion. □

Driessen et al. (2012) justify this inception-based estimator under a different asymptotic device than the one used here. Their Theorem 1 fixes the number of vintage-year portfolios and lets the number of underlying projects or funds within each portfolio diverge, yielding consistency for the inception-only estimator under their maintained model (Driessen et al., 2012, p. 520). Economically, this portfolio-infill asymptotic sequence averages out within-portfolio idiosyncratic noise, but it does not by itself create new realizations of aggregate states. Relative to that setup, our framework preserves the exact $H = 0$ nesting result while letting the vintage-year domain itself expand, which is better aligned with unconditional identification, horizon comparisons, and overlap dependence when new vintages bring new realizations of macroeconomic conditions.

1.4.2 Connection to Driessen et al. (2009)

The earlier working-paper version, Driessen et al. (2009), provides the closest conceptual precursor to our NFV approach. Starting from the same inception-based NPV criterion as Driessen et al. (2012), they show that the raw sum of squared NPVs is not globally convex in finite samples (the "exploding alpha" problem mentioned before) and stabilize estimation by transforming the NPV relation itself. Their two proposed "GMM-style" criteria are a log-transformed specification based on the difference between the log present values of investments and dividends and a PME-style ratio specification akin to Kaplan and Schoar (2005) that minimizes the distance between the ratio of these present values and one. This is the main conceptual connection to our framework: both approaches use transformed-NPV criteria to obtain a better-behaved objective, but the source of the resulting bias is different.

To make the asymptotic logic explicit without introducing additional counting notation, let $\overline{PV}^D(\theta)$ and $\overline{PV}^I(\theta)$ denote the discounted portfolio averages of dividends and investments within a generic vintage-year portfolio, and let $\overline{NPV}(\theta) = \overline{PV}^D(\theta) - \overline{PV}^I(\theta)$. Under the working paper's asymptotic scheme, the number of funds inside each fixed vintage-year portfolio diverges, so the

PME transformation inherits the same population zero set as the inception-NPV criterion:

$$\frac{\mathbb{E}[\overline{PV}^{D}(\theta_0)]}{\mathbb{E}[\overline{PV}^{I}(\theta_0)]} - 1 = \frac{\mathbb{E}[\overline{NPV}(\theta_0)]}{\mathbb{E}[\overline{PV}^{I}(\theta_0)]}.$$

Hence, when $\mathbb{E}[\overline{NPV}(\theta_0)] = 0$ and $\mathbb{E}[\overline{PV}^{I}(\theta_0)] > 0$, the transformed and untransformed portfolio criteria are asymptotically equivalent. The cost of stabilization is therefore not a different asymptotic target, but a finite-sample nonlinearity bias: Driessen et al. (2009) find a downward bias for α under the log transformation and an upward bias under the ratio specification.

The contrast to our framework is that their transformation is applied after within-portfolio averaging. If the ratio transformation were treated as primitive at the unit level before aggregation, the target would change. For a generic unit i with $NPV_i(\theta) = PV_i^D(\theta) - PV_i^I(\theta)$, one obtains at θ_0

$$\mathbb{E}\left[\frac{PV_i^D(\theta_0)}{PV_i^I(\theta_0)}\right] - 1 = \mathbb{E}\left[\frac{NPV_i(\theta_0)}{PV_i^I(\theta_0)}\right] = \mathrm{Cov}\left(NPV_i(\theta_0), \frac{1}{PV_i^I(\theta_0)}\right),$$

where the last equality uses $\mathbb{E}[NPV_i(\theta_0)] = 0$. This expectation of a random ratio is not the ratio-of-expectations moment justified by Driessen et al. (2009). The Driessen et al. (2009) construction therefore stabilizes estimation while preserving the same large-sample target under portfolio-infill asymptotics, whereas in our setup changing H changes the set of discounting dates and hence the population criterion itself. Different horizons can therefore target different pseudo-true parameters $\theta_H^\dagger$ even asymptotically. A practical advantage of our horizon-based approach is that it operates directly on net cash-flow streams. The two alternatives in Driessen et al. (2009), by contrast, require separately observed and accurately labeled investment and dividend cash flows.

1.4.3 Connection to Korteweg and Nagel (2016)

Our framework is close to Korteweg and Nagel (2016) at the level of primitive discounted cash-flow moments, but it does not literally nest their estimator. If the cross-sectional unit is allowed to be an arbitrary pooled portfolio or calendar-time window, our notation can represent the same

private-equity cash-flow moment building blocks that enter their GMM setup. In that sense, the connection is economically close.

The main difference is the sample objective. Our LMD estimator applies the loss before aggregation, minimizing an average of unit-level losses, whereas Korteweg and Nagel (2016) apply a GMM quadratic form to an aggregated mean moment vector. While both approaches target the true parameter θ_0 under correct model specification, this distinction becomes economically meaningful under misspecification. Because the *average of losses* applies a different implicit asset weighting than the *loss of an average*, the two approaches will generally converge to different pseudo-true parameter vectors when the model is misspecified, a pervasive feature of asset pricing models. In addition, their empirical implementation assigns zero weight to the private-equity pricing-error block during estimation, so the SDF is pinned down by public-equity and T-bill moments and the PE pricing error is evaluated only afterwards. Accordingly, Korteweg and Nagel (2016) are best viewed as a closely related aggregated-moment GMM analogue rather than as a special case of Equation (13).

The asymptotic difference mirrors this design choice. The Korteweg and Nagel (2016) approach is natural when identification comes from repeated aggregate-state realizations: enlarging the cross section within a common payoff window averages out idiosyncratic fund noise, but not the common SDF realization itself. Their asymptotic logic therefore relies on additional nonoverlapping payoff windows rather than on cross-sectional replication alone (Korteweg and Nagel, 2016, p. 1447). Our framework instead uses increasing-domain asymptotics indexed by vintage years, with bounded within-vintage counts and bounded fund lifetimes, so that new vintages contribute new macroeconomic realizations while dependence remains local.

1.4.4 Positioning of the present estimator

The present estimator is best viewed as a synthesis tailored to the current application. It combines the exact inception-only objective link to Driessen et al. (2012), the stabilization concern emphasized in Driessen et al. (2009), and the Korteweg and Nagel (2016) insight that identification requires

new aggregate-state realizations rather than cross-sectional replication alone. Operationally, these elements enter a pooled-unit nonlinear least-squares framework with explicit horizon control.

The practical distinction is which form of sample expansion supplies new identifying information. If a researcher mainly gains more funds inside already-observed vintages, the Driessen et al. (2012) infill logic is the natural approximation. If the sample mainly lengthens in calendar time and thereby adds nonoverlapping payoff windows, the Korteweg and Nagel (2016) logic is closer to the underlying information structure. If the panel broadens chiefly by observing additional vintages while per-vintage counts remain modest, our increasing-domain framework is the more natural approximation. Table 1 summarizes the main distinctions between Driessen et al. (2012), Korteweg and Nagel (2016), and our approach.

2 Implementation choices and simulation

In this section and the next, we move from the general methodology to our empirical application. First, we describe the data and establish our sample restriction based on terminal cash-flow measurement concerns. Second, we detail the empirical implementation choices that anchor the general estimator from Section 1.2. Third, we use a simulation study to validate and calibrate these choices. Finally, Section 3 presents the resulting empirical estimates for Buyout and Venture Capital funds.

2.1 Private data

We use the Preqin cash flow dataset as of 14th April 2022, which is well known in the academic private equity literature (Harris et al., 2014; Brown et al., 2015; Korteweg and Nagel, 2016; Ang et al., 2018). To keep the empirical analysis clear and concise, we filter the dataset for Buyout (BO) and Venture Capital (VC) funds. We only keep funds with fund currency US Dollar (USD) and region equal to North America.

For this subset of BO and VC funds, we extract all (i) equal-weighted (hereafter EW) and (ii) value-weighted (hereafter FW, weighted by fund size) cash flow series (cf. Figures 3 and 4). In

the overall dataset, the minimum vintage year is 1982, and the maximum is 2021. We initially retain all vintage years, including recent ones. Section 2.3 assesses the impact of unseasoned fund Net Asset Values (NAVs) on our empirical estimation results. On this basis, we decide how many recent vintage years to remove to reduce reliance on unrealized NAVs. For the vintage years we keep, all liquidated and non-liquidated funds enter our estimation. For non-liquidated funds, we treat the latest NAV as final cash flow.

Buyout sample. The Buyout (North America) sample contains 894 distinct funds spreading over 36 vintage years. The vintage year distribution is as follows: 1985 (2), 1986 (3), 1987 (5), 1988 (6), 1990 (2), 1991 (3), 1992 (6), 1993 (4), 1994 (14), 1995 (11), 1996 (13), 1997 (14), 1998 (29), 1999 (24), 2000 (34), 2001 (19), 2002 (13), 2003 (19), 2004 (24), 2005 (43), 2006 (48), 2007 (44), 2008 (43), 2009 (19), 2010 (31), 2011 (24), 2012 (36), 2013 (34), 2014 (44), 2015 (34), 2016 (54), 2017 (35), 2018 (43), 2019 (63), 2020 (32), 2021 (22). The region distribution is as follows: North America (461), US (433). The strategy distribution is as follows: Buyout (894). The fund currency distribution is as follows: USD (894).

Venture Capital sample. The Venture Capital (North America) sample contains 931 distinct funds spreading over 38 vintage years. The vintage year distribution is as follows: 1982 (1), 1985 (3), 1986 (4), 1987 (2), 1988 (1), 1989 (1), 1990 (4), 1991 (1), 1992 (7), 1993 (6), 1994 (6), 1995 (8), 1996 (14), 1997 (15), 1998 (28), 1999 (44), 2000 (68), 2001 (45), 2002 (26), 2003 (18), 2004 (31), 2005 (34), 2006 (43), 2007 (46), 2008 (36), 2009 (18), 2010 (18), 2011 (22), 2012 (20), 2013 (32), 2014 (31), 2015 (37), 2016 (34), 2017 (48), 2018 (46), 2019 (59), 2020 (44), 2021 (30). The region distribution is as follows: North America (267), US (664). The strategy distribution is as follows: Early Stage (268), Early Stage: Seed (44), Early Stage: Start-up (23), Expansion / Late Stage (123), Venture (General) (451), Venture Debt (22). The fund currency distribution is as follows: USD (931).

Typical PE funds call approximately 33% of their total capital commitments by year one,

50% by year two, 66% by year three, 80% by year four, 88% by year five, and 93% by year six. These cumulative drawdown dynamics clarify why short compounding horizons come with smaller structural target shifts. Because investments initiated *after* the maximum evaluation date in $\mathcal{T}_{i,H}$ are priced strictly as future projects ($\tau < t_{i,j}^{\text{Inv}}$), their conditional expected pricing error remains zero. Therefore, they are mathematically immune to the Compounding-Horizon Wedge. For example, under a compounding horizon of $H = 4$ years, roughly 20% of the aggregate investment cash flows occur *after* the active averaging window. Thus, for $H = 4$ roughly 20% of the fund's capital is discounted without accumulating the covariance wedge that structurally drives the target shift.

2.2 SDF models and public data

We use as our baseline main-text specification a simple linear SDF model similar to Driessen et al. (2012)

$$\Psi_{\tau,t}^{\text{SL}}(\theta) = \frac{\prod_{h=1}^{\tau}\left(1 + \alpha + r_{f,h} + \sum_m \beta_m F_{m,h}\right)}{\prod_{h=1}^{t}\left(1 + \alpha + r_{f,h} + \sum_m \beta_m F_{m,h}\right)} \tag{19}$$

with monthly (arithmetic) risk-free return r_f, monthly (arithmetic) zero-net-investment portfolio returns F_m, and parameter vector $\theta = (\alpha, \beta)$. Throughout both the simulation and the empirical application, all factor returns and fund cash flows are observed at monthly frequency.

The public market factors that enter our SDF draw on the US dataset of the recently popularized q^5 investment factor model sourced from `http://global-q.org/factors.html` (Hou et al., 2015, 2021). According to Hou et al. (2019) this family of new "investment" factors subsumes the more popular Fama-French factors (Fama and French, 2015, 2018), which indicates higher explanatory power of the Hou et al. (2021) model. The q^5 five-factor model includes the market excess return (MKT), a size factor (ME), an investment factor (IA), a return on equity factor (ROE), and an expected growth factor (EG). The maximum date in the q^5 investment factor dataset is 2024-12-31.

To avoid overfitting, we restrict our analysis to six simple SDF models that contain {MKT} alone or {MKT} plus {ME or IA or ROE or EG or Alpha}. In Equation (13), we use the quadratic

loss function $L(x) = x^2$.

In our optimization procedure, we limit the alpha intercept α to the range $[-2\%, +2\%]$ per month to avoid the "exploding alpha" issue documented in (Driessen et al., 2012, Footnote 8): when α is unrestricted, an arbitrarily large α value can almost perfectly price any cash flow stream (except its first component), rendering the beta risk factors irrelevant.

2.3 The impact of unseasoned funds and reported NAVs

Before defining our baseline sample, we highlight a structural limitation of including recent vintages: treating the final reported NAV as a terminal cash flow substitutes an expected continuation value for a realized payoff, violating the actual-divestment requirement of Assumption 1. We mitigate this by imposing a strict maximum vintage-year cutoff to ensure the sample primarily consists of seasoned funds with minimal remaining NAVs.

Figure 5 presents the corresponding sensitivity analysis for BO and VC under fund-value weighting (FW). In both sub-samples, the estimated market beta drifts upward monotonically as more recent vintages are included. For BO, $\hat{\beta}_{\text{MKT}}$ at the 0-year Horizon (see Remark 1) rises from approximately 1.23 with a 2011 cutoff to 2.42 with the full sample (2021), with most of the increase concentrated after 2016. For VC, the FW estimate climbs from 1.57 to 2.33 over the same range. Discounting all final NAVs by 50% substantially flattens this trend: the BO FW range compresses to 1.09–1.65 and the VC FW range to 1.41–1.54, consistent with the view that the drift is driven by mechanical reliance on subjectively reported NAVs for young funds. These findings underpin the sample restriction to the 2010 maximum vintage year used throughout. Figure 6 confirms that unseasoned post-2016 vintages similarly inflate the estimated intercept α in two-factor models, an effect mitigated by the 50% NAV discount.

Remark 1 (Definition of Horizon)**.** *Throughout the empirical section, we use the term* Horizon *H to denote the maximum compounding horizon in years used in the set of discounting dates $\mathcal{T}$ (cf. Equation* (12)*). Formally, starting from the fund inception date t_i^0, estimating a model with a Horizon H implies that the estimator minimizes the average pricing error over the set of discounting*

dates $\mathcal{T}_{i,H} = \{t_i^0, t_i^0 + 1, \ldots, \min(t_i^0 + 12H, T_i^{\max})\}$*, formally denoted as* $\bar{\epsilon}_i^{(H)}$*. Note that the Horizon has as upper bound the max cash flow date per fund or portfolio in our empirical application. This Horizon in years serves as x-axis in all subsequent empirical and simulation charts.*

2.4 NAV discount sensitivity analysis

To further quantify the sensitivity of factor loading estimates to terminal NAV treatment, we conduct a systematic NAV discount analysis on the seasoned sample (maximum vintage year 2010). Figure 7 reports single-factor MKT estimates under five NAV treatments: full reported NAV, 25%, 50%, 75% constant discounts, and a complete write-off.

For BO, $\hat{\beta}_{\text{MKT}}$ decreases monotonically with the discount. At the 0-year Horizon under EW, estimates range from 1.26 (no discount) to 0.94 (100% write-off); under FW, from 1.20 to 0.98. At the 15-year Horizon, the EW range is 0.81–0.55 and the FW range 0.84–0.62. The single-factor market model remains economically plausible across all treatments.

For VC, the pattern is analogous but the sensitivity is more muted under EW (0-year range: 1.88–1.74) due to the higher baseline beta. Under FW, the effect is more pronounced, particularly at longer Horizons (15-year range: 1.07–0.73).

The monotonic relationship is consistent with reported NAVs mechanically inflating betas, as subjective valuations co-move with public indices more than realized exits do. Even under a complete write-off, BO maintains a market beta near or below unity while VC retains substantially higher systematic exposure. Rather than imposing a model-based residual-value adjustment (Jenkinson et al., 2020; Brown et al., 2023), we use the range spanned by the five treatments as a transparent envelope of estimation uncertainty attributable to terminal value measurement.

2.5 Simulation study

Our Monte Carlo experiments examine the finite-sample bias and variance of the estimator. By construction, these simulations assess the estimator's ability to recover the parameters of an assumed data-generating process, rather than the empirical validity of the underlying SDF model itself.

Specifically, we investigate (i) the efficiency gains of vintage-year portfolios versus individual funds; (ii) the tradeoff between the time-series (V) and cross-sectional (n/V) dimensions; (iii) multivariate identification under idiosyncratic noise; and (iv) the optimal choice of the maximum compounding Horizon $\mathcal{T}$.

The baseline simulation uses historical q-investment factor paths from 1986 to 2005 and generates 20 funds per vintage year (V=20). Subsequent paragraphs vary one margin at a time; full scenario definitions are in the online repository. The "extended vintage" scenarios use vintages 1967–2005 (V=39). Each fund contains 15 deals with equal investment amounts and one divestment cash flow. Deals are entered within the first five years (discrete uniform) and held for one to ten years (uniform). Returns follow the linear SDF of Equation (19) with β_{MKT}=1, augmented by a normal i.i.d. monthly error (σ=0.2, zero mean); negative returns exceeding −100% trigger default with zero exit. Unlike the shifted lognormal error of Driessen et al. (2012), which precludes returns below roughly −99%, our specification admits full default; we analyze the remaining design divergence in Appendix D.1. We evaluate the estimator over Horizons from 0 to 30 years and simulate 1000 replications for both vintage-year and single-fund portfolios.

Cross-sectional unit i: As anticipated in Subsection 1.3, vintage-year portfolios exhibit lower bias and variance than individual funds. At a 15-year Horizon, the mean (standard deviation) of $\hat{\beta}_{\text{MKT}}$ is 0.966 (0.188) for vintage-year portfolios versus 0.887 (0.359) for individual funds. Figure 8 illustrates that this advantage extends across the full Horizon curve: single-fund estimates are severely upward-biased at short Horizons ($\hat{\beta}_{\text{MKT}} \approx 1.5$) and deteriorate to negative values at long Horizons, whereas vintage-year portfolios remain in a narrow range around 1.0.

This bias–variance improvement comes at the cost of dramatically fewer cross-sectional units, weakening asymptotic approximations. This constraint likely explains why Korteweg and Nagel (2016) use individual funds in their SHAC framework, exploiting a larger sample for smaller standard errors.

Varying vintages V and portfolio sizes n/V: A key question is whether precision improves more from extending V or from enlarging n/V. For this comparison, we switch to a reference scenario with V=20, n/V=40 and compare it with four alternatives that vary V and n/V (see Figures 10–9).

Increasing n/V is substantially more effective at suppressing variance than adding vintages. At the 15-year Horizon, doubling n/V from 10 to 20 over 39 years reduces $\sigma(\hat{\beta}_{\mathrm{MKT}})$ by roughly 22% (0.218 to 0.169). Shrinking V from 39 to 10 years elevates variance only moderately for the early sub-sample (1986–1995, $\sigma \approx 0.194$), though the late sub-sample (1996–2005) exhibits considerably higher variance ($\sigma \approx 0.432$), reflecting macro-environment sensitivity. This hierarchy is even more pronounced in the two-factor setting, where increasing n/V aggressively dampens spurious α variance. These findings provide an empirical rationale for the finite-sample superiority of Driessen et al. (2012) infill asymptotics (fixing V, $n_v \to \infty$) over purely time-series aggregation.

Two-factor models (q^5 factors): Figure 11 reports two-factor simulations pairing MKT with each non-market q^5 factor (ME, IA, ROE, EG), with β_{MKT}=1 and β_{2nd}=0.5. A common pattern emerges: the second-factor loading is overestimated at short Horizons and attenuated at long Horizons, while MKT absorbs a compensating bias. Variance is markedly larger than in the single-factor case due to factor collinearity.

The ME model is the only specification exhibiting a sign change in $\hat{\beta}_{\mathrm{2nd}}$ (at ~17.5 years). The IA model shows large short-Horizon bias ($\hat{\beta}_{\mathrm{IA}} \approx 1.06$) but converges near the true values around 17.5 years. The EG model has the highest variance ($\sigma(\hat{\beta}_{\mathrm{EG}}) \approx 1.5$ at Horizon 0) but achieves reasonable accuracy around 15–17.5 years. The ROE model is the exception: both loadings remain persistently biased at all Horizons. These results counsel caution when interpreting two-factor SDF estimates and indicate that the MSE-minimizing Horizon is model-dependent, typically in the 12.5–17.5-year range.

Varying fund lifetimes: Figure 12 illustrates how the bias-minimizing Horizon shifts with fund lifetime. For the shortest examined lifetime (one-year investment, one-year hold), beta bias is minimized at a 12.5-year Horizon and alpha bias at 5 years; for a two-year/two-year fund, the

corresponding optima are 15 and 7.5 years. These results show that the bias-minimizing Horizon can extend well beyond a fund's actual lifetime.

Size of set $\mathcal{T}$ (Maximum Compounding Horizon): A central finding concerns the choice of $\mathcal{T}$. Theoretically, discounting exclusively to inception (H=0), i.e. the Driessen et al. (2012) framework, yields unbiased estimates under the exact-pricing restriction (Equation (1)). However, in the reported simulation designs, bias is minimized not at H=0 but around 12.5–15 years: at short Horizons, idiosyncratic noise interacts non-linearly with estimation, inducing severe factor-loading bias. Averaging pricing errors over multiple dates acts as a finite-sample stabilization device, improving the mean estimate far more than it reduces standard errors, while introducing the target shift mentioned in Section 1.2. Our shorter-lifetime simulations indicate that the optimal horizon requires tailored calibration rather than a simple fund-life heuristic. But, because the finite-sample gains from horizon averaging are DGP-dependent and come with the horizon-specific target shift, horizon choice should at present be viewed as a tuning decision guided by simulations, stability diagnostics, and economic judgment rather than by a fully developed optimality theory. We further analyze these results theoretically in Appendix B.

Simulation conclusion: To conclude, our simulation study supports several implementation choices used in the empirical application. First, vintage-year portfolio formation helps improve estimator precision relative to single-fund estimation. Second, in the reported designs, increasing the number of funds per vintage appears more effective in controlling estimator variance than increasing the number of vintages (for i.i.d. error terms). Third, averaging pricing errors over multiple dates materially reduces the short-Horizon bias of the inception-only criterion in the reported designs, even though Section 1.1 shows that this stabilization comes with a horizon-specific target shift.

Accordingly, in the empirical application we use vintage-year portfolios and the simple linear SDF as the baseline specification, emphasize one-factor MKT specifications, treat two-factor results as exploratory, and focus on Horizons around 12.5–15 years while also reporting sensitivity over a

broader grid.

2.6 Role of asymptotic inference

We compute asymptotic standard errors using the SHAC framework of Subsection 1.3 with Bartlett kernel bandwidth b_V=12 years. These provide the benchmark large-sample approximation.

Because private-equity samples remain small and model choice is often data-driven, we complement asymptotic inference with hv-block cross-validation (Racine, 2000), a consistent model-selection criterion for dependent data. Each fold partitions the data into three disjoint groups: (i) a *validation set* (v-block) of three consecutive vintage years; (ii) a *buffer zone* (h-block) of three vintage years on each side, discarded to break dependence from overlapping cash flows; and (iii) an *estimation set* comprising all remaining vintage years. This three-way split is repeated across rolling vintage-year windows (Table 2), with the number of active folds adapting dynamically to the maximum vintage-year cutoff (e.g., seven folds when the sample is truncated at 2010).

Remark 2 (Cross-validation as stability diagnostic)**.** *The hv-block cross-validation quantities reported throughout this paper serve as* ***stability diagnostics and model-selection criteria*** *rather than inferential standard errors. They summarize the fold-to-fold variation in parameter estimates under dependence-reducing sample splits and are informative about model sensitivity and effective parameter curvature, but they do not provide a valid null distribution for the full-sample estimator. For formal statistical inference (standard errors and hypothesis tests), we rely on the SHAC-based asymptotic benchmark of Section 1.3. In practice, hv-block cross-validation complements the asymptotic analysis by (i) diagnosing model stability and out-of-sample performance without relying on asymptotics and (ii) flagging over-parameterized specifications that appear well-identified in-sample but collapse out-of-sample.*

3 Empirical results for BO and VC data

This section presents the empirical SDF estimates for Buyout (BO) and Venture Capital (VC) funds and uses them to assess which specifications are stably identified in the current cash-flow panel. As the simulation study in Subsection 2.5 confirms, we use vintage-year portfolios (VYPs) to estimate simple linear SDF models for the BO and VC sub-samples individually. We restrict the data to North American, USD-denominated funds from the Preqin dataset described in Subsection 2.1. This restriction ensures consistency between fund cash flows and the U.S. q-investment factors used throughout the analysis. Moreover, we only consider funds with a maximum vintage year of 2010, to reduce the reliance on unrealized NAVs (see Section 2.3).

We analyze BO and VC as two distinct asset classes with fundamentally different risk profiles and report both equal-weighted (EW) and value-weighted (FW) estimates across a grid of compounding Horizons. Our baseline specification is the single-factor MKT model estimated on vintage-year portfolios. Two-factor models that add an alpha intercept or a single q-investment factor serve as robustness and exploratory extensions.

For each model specification, we report asymptotic standard errors (SE_A) from the SHAC framework alongside hv-block cross-validation fold dispersions (SD_{CV}), defined as the standard deviation of parameter estimates across hold-out folds. Throughout the empirical analysis, t_A denotes the t-ratio based on asymptotic standard errors; SR_{CV} denotes the signed stability ratio $\mathrm{SR}_{CV} := \hat{\theta}/\mathrm{SD}_{CV}$, where $\hat{\theta}$ is the corresponding full-sample estimate. Because SR_{CV} preserves the sign of $\hat{\theta}$, negative values simply indicate negative estimated coefficients; its absolute value $|\mathrm{SR}_{CV}|$ measures the coefficient magnitude relative to fold-to-fold dispersion. We use SR_{CV} descriptively rather than as a formal test statistic. As discussed in Remark 2, SR_{CV} complements asymptotic inference as a stability diagnostic. We report asymptotic significance markers for completeness, but when SE_A is erratic, boundary solutions arise, or the cross-validation diagnostics indicate instability, we do not treat isolated significant t_A values as decisive.

All R code and data (including the previous simulations) are available in an online repository: `https://github.com/quant-unit/Fundwise_SDF/tree/master/r_project`.

3.1 Baseline: single market-factor models

Figure 13 and Tables 3–4 report the single-factor MKT model for BO and VC across Horizons from 0 to 30 years. As anticipated by the simulation study, the market beta declines with the compounding Horizon for both asset classes.

For BO under EW, $\hat{\beta}_{\text{MKT}}$ ranges from approximately 1.20–1.24 at short Horizons (0–5 years) and decreases to 0.68 at 25–30 years; under FW, the corresponding range is 1.22–1.23 down to 0.71. The divergence between asymptotic and cross-validation diagnostics persists: asymptotic t-ratios (t_A) are moderate at short Horizons (e.g., 0.94–1.05 under EW at 0–5 years) and decline further at longer Horizons, while the cross-validation stability ratios remain high throughout (SR_{CV} ranging from 4.98 to 10.01 under EW and from 2.59 to 8.09 under FW), indicating comparatively stable estimates across data subsets.

For VC, the market beta is substantially higher across all Horizons, consistent with venture capital's greater sensitivity to systematic risk. Under EW, $\hat{\beta}_{\text{MKT}}$ reaches approximately 1.97–2.00 at 2.5–5 years and converges toward 1.05–1.18 beyond 25 years; under FW, the corresponding range is 1.66 down to 0.82–0.90. Despite the higher parameter variability reflected in larger asymptotic standard errors (SE_A of 1.28–1.83), the cross-validation stability diagnostics indicate comparatively stable estimates (SR_{CV} of 3.65–4.55 under EW; 3.53–7.57 under FW), although somewhat lower than the BO stability ratios due to the more heterogeneous VC fund universe. As horizon-specific diagnostics rather than direct structural estimates, the converged long-horizon coefficients (BO ≈ 0.68, VC ≈ 1.1 under EW) remain economically interpretable: BO funds, which invest in mature companies with stable cash flows, load less on market risk than VC funds, which back early-stage ventures with highly uncertain outcomes.

3.2 Robustness: two-factor $\alpha + MKT$ models

Figure 14 and Tables 5–6 present the two-factor MKT + α model for BO and VC. The results show the instability anticipated by the simulation study: once an alpha intercept is introduced, separation between the intercept and the MKT loading weakens materially.

For BO, the alpha term absorbs a substantial share of the variation that otherwise loads on the market factor: under EW, $\hat{\alpha}$ ranges from 0.003 to 0.006 per month (and -0.001 for 17.5 years) and the MKT coefficient oscillates between 0.35 and 1.01, with no stable convergence across Horizons. Under FW, $\hat{\alpha}$ is persistently around 0.004–0.008 per month while the MKT loading falls to values between 0.11 and 0.44. The asymptotic standard errors for FW are highly erratic, with several entries at or very near zero interspersed with values exceeding 25. Cross-validation fold dispersions for $\hat{\beta}_{\text{MKT}}$ are large (0.16–0.44 under EW; 0.64–0.74 under FW), indicating that these point estimates are highly sensitive to the sample split. Accordingly, the few isolated asymptotic significance markers in Tables 5–6 are not emphasized in our interpretation.

For VC, the two-factor model exhibits a different but similarly unstable pattern. At short Horizons (0 under EW), the alpha term reaches its upper bound ($\hat{\alpha}$ = 0.020/month) and the MKT coefficient drops to approximately 0.47, resembling the BO behavior. However, as the Horizon increases beyond 5 years, the MKT loading rises steadily to values exceeding 2.3, while the alpha estimate turns progressively negative, reaching −0.009/month at 30 years. We interpret this reversal as further evidence that the two-parameter decomposition is highly sensitive to Horizon choice and sample composition in VC. Under FW, the instability is somewhat attenuated ($\hat{\alpha}$ remains close to zero at short Horizons; $\hat{\beta}_{\text{MKT}} \approx 1.74$), but the overall fragility persists, with asymptotic standard errors exceeding 10 for some Horizons.

The BO and VC two-factor results therefore reinforce the same conclusion: among the specifications considered here, single-factor MKT models are the most stable for private-equity SDF estimation in our sample.

3.3 Robustness: two-factor q^5-factor models

We now pair MKT with each of the four non-market q^5 factors (ME, IA, ROE, EG) in two-factor specifications estimated separately for BO and VC. Because the detailed estimation tables contain extensive Horizon-by-Horizon results, we summarize only the main patterns visible in Figures 15 and 16; the full tables are available in the accompanying online repository.

For BO, the qualitative message is similar across the four non-market factors. Second-factor loadings are often positive at short Horizons, but with increasing Horizon they generally attenuate, fluctuate around zero, or change sign. ME is the strongest partial exception, yet even there the loading is not stable across the full Horizon grid and weighting schemes. By contrast, the MKT loading remains close to the corresponding single-factor estimate, so adding a second q^5 factor does not materially change the market-risk assessment. At the same time, the asymptotic standard errors are frequently erratic and sometimes extremely large, which reinforces the limited reliability of asymptotic inference in this setting.

VC exhibits the same pattern more starkly. Several second-factor loadings are very large at short Horizons, but they vary sharply with the compounding Horizon and often reverse sign at intermediate or long Horizons. This degree of Horizon sensitivity closely matches the weak-identification patterns documented in the simulation study. Across specifications, the resulting estimates are materially less stable and less economically interpretable than the single-factor MKT benchmark.

A few second-factor loadings attain conventional asymptotic significance at isolated Horizons, but those signals are neither persistent across the Horizon grid nor matched by comparably strong cross-validation evidence. Overall, none of the four non-market q^5 factors yields a stable, economically interpretable loading across Horizons for either asset class. We therefore do not view these two-factor specifications as providing evidence as stable as the single-factor MKT model in the present sample.

4 Conclusion

Our generalized Least-Mean-Distance framework provides a flexible way to study identification and estimation of stochastic discount factors using non-traded cash flows. Methodologically, the paper recasts semiparametric pooled-cash-flow pricing as a nonlinear LMD problem with flexible discounting dates, nests the inception-only criterion of Driessen et al. (2012), and clarifies why closely

related horizon-averaged procedures need not target the same population object. The central theoretical message is that Horizon choice is part of the estimand: inception-only discounting preserves the structural pooled pricing restriction, whereas averaging pricing errors over multiple dates can improve finite-sample recoverability (through both lower dispersion and lower higher-order bias) while generally shifting the population target to a Horizon-specific pseudo-true parameter. Our increasing-domain asymptotic framework for overlapping vintages provides a benchmark for this identification problem. Taken together, the methodological results answer the paper's opening question: with pooled non-traded cash flows, what can be identified with stability depends jointly on factor specification, the cash-flow data-generating process, and the chosen compounding Horizon. Our Monte Carlo results show that the tradeoff between structural fidelity and finite-sample recoverability is economically relevant in the reported designs, with short-Horizon criteria often substantially less stable than intermediate-Horizon averages, especially once abnormal performance (α) is estimated jointly with market exposure.

In the private-equity application, the main empirical takeaway is diagnostic rather than definitive. Current seasoned cash-flow panels appear informative about coarse market exposure in BO and VC portfolios, but much less informative about abnormal performance and richer factor decompositions. Across our specifications, the single-factor MKT model is the most stable, whereas adding α or a second non-market factor makes the estimates markedly more sensitive to Horizon choice and often weakens their economic interpretability. We do not read this as evidence that private equity is priced by a single factor in population or that private-equity alpha is absent. Rather, we read it as evidence that, under currently available pooled cash-flow data and semiparametric estimation, multi-parameter SDF models are only weakly identified in the present sample. Accordingly, empirical practice in this setting should emphasize parsimonious specifications, robustness over a Horizon grid, and dependence-robust stability diagnostics such as hv-block cross-validation, while treating asymptotic significance as a useful benchmark rather than decisive finite-sample evidence.

More broadly, our results suggest that progress on richer private-equity SDF models will require either more informative data or additional structure. Promising directions include better

integration of interim valuation information and cash-flow dynamics, regularization or model-selection methods for sparse multi-factor settings, and principled procedures for choosing the set of discounting dates $\mathcal{T}$. These extensions would help clarify when Horizon averaging improves finite-sample performance enough to justify the associated change in estimand.

List of Figures

1 **Interpretation of averaged pricing error.** How to calculate and interpret $\mathcal{T}$-averaging introduced by Equation (11): The time index t is relevant for the cash flows (red and green dots). The time index τ is used for the pricing error, i.e., the sum of discounted net cash flows (blue boxes). The weighted average of these "pricing errors" gives the average pricing error $\bar{\epsilon}$ as defined in Equation (11) (solid blue line). In this example, card($\mathcal{T}_i$) = 7, i.e., the number of blue boxes. 51

2 **Regularizing the sample objective function.** The figure plots the sample objective $Q_{n,H}(\theta)$ with quadratic loss over α and β_{MKT}, using simulated cash flows from the base-case DGP (Section 2.5). The green dot marks the true parameters $(\alpha_0, \beta_{\text{MKT},0}) = (0, 1)$. Left panel ($H = 0$): The inception-only criterion suffers from the "exploding alpha" degeneracy. Large positive α values drive multiplicative discount factors to zero, creating a ridge of spurious near-minima along the α axis that prevents reliable identification. Right panel ($H = 15$ years): Horizon averaging regularizes the objective surface, restoring a well-identified minimum near the true parameters. 52

3 **Cumulative cash flows of BO funds.** J-curves by vintage year for BO funds in Preqin dataset scaled by fund commitment. Post-2010 vintage years are shown as green lines. Based on the sensitivity analysis in Section 2.3, these vintages are excluded from the empirical estimates in Section 3 to limit reliance on unrealized NAVs. 53

4 **Cumulative cash flows of VC funds.** J-curves by vintage year for VC funds in Preqin dataset scaled by fund commitment. Post-2010 vintage years are shown as green lines. Based on the sensitivity analysis in Section 2.3, these vintages are excluded from the empirical estimates in Section 3 to limit reliance on unrealized NAVs. 54

5 **Sensitivity analysis (MKT factor).** Single-factor model for BO (left) and VC (right) under value weighting, using VYP with different vintage-year cutoffs (asymptotic estimates). For each asset class, the left chart shows the standard baseline (full NAV as final cash flow), while the right chart applies a 50% discount to all final NAVs. 55

6 **Sensitivity analysis (α + MKT factors).** Two-factor model for BO (left) and VC (right) under value weighting, using VYP with different vintage-year cutoffs (asymptotic estimates). For each asset class, the left chart shows the standard baseline (full NAV as final cash flow), while the right chart applies a 50% discount to all final NAVs. 56

7 **NAV discount sensitivity (MKT factor).** Single-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010. Each panel shows asymptotic estimates under five NAV treatments: full NAV as final cash flow, 25%, 50%, 75% constant discounts, and 100% NAV write-off. 57

8 **Simulation results (α + MKT factors).** Single-fund vs. vintage-year portfolios. . . 58

9 **Simulation results (MKT factor).** Varying vintages V and portfolio sizes n/V. . . 59

10 **Simulation results (α + MKT factors).** Varying vintages V and portfolio sizes n/V. 60

11 **Simulation results (q^5 factors).** Two-factor models using VYP. 61

12 **Simulation results (α + MKT factors).** Varying fund lifetimes using VYP with the following yearly investment-period/holding-period pairs: 1/1, 2/2, 3/3, 3/6, and 3/10 (from left to right). 62

13 **Empirical results for BO and VC (MKT factor).** Single-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010. 63

14 **Empirical results for BO and VC (α + MKT factors).** Two-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010. 64

15 **Empirical results for BO and VC (q^5-factors).** Two-factor model for BO funds using VYP and maximum vintage year 2010. 65

16 **Empirical results for BO and VC (q^5-factors).** Two-factor model for VC funds using VYP and maximum vintage year 2010. 66

References

Andonov, A., Kräussl, R., and Rauh, J. (2021). Institutional investors and infrastructure investing. *The Review of Financial Studies*, 34(8):3880–3934.

Ang, A., Chen, B., Goetzmann, W. N., and Phalippou, L. (2018). Estimating private equity returns from limited partner cash flows. *Journal of Finance*, 73(4):1751–1783.

Bao, Y. and Ullah, A. (2007). Finite sample properties of maximum likelihood estimator in spatial models. *Journal of Econometrics*, 137(2):396–413.

Boyer, B. H., Nadauld, T. D., Vorkink, K. P., and Weisbach, M. S. (2023). Discount-rate risk in private equity: Evidence from secondary market transactions. *The Journal of Finance*, 78(2):835–885.

Brown, G. W., Ghysels, E., and Gredil, O. R. (2023). Nowcasting net asset values: The case of private equity. *The Review of Financial Studies*, 36(3):945–986.

Brown, G. W., Gredil, O. R., and Kaplan, S. N. (2019). Do private equity funds manipulate reported returns? *Journal of Financial Economics*, 132(2):267–297.

Brown, G. W., Harris, R. S., Jenkinson, T., Kaplan, S. N., and Robinson, D. T. (2015). What do different commercial data sets tell us about private equity performance? Available at SSRN 2706556.

Burnside, C. (2016). Identification and inference in linear stochastic discount factor models with excess returns. *Jnl of Financial Econometrics*, 14(2):295–330.

Cordell, L., Roberts, M. R., and Schwert, M. (2023). Clo performance. *The Journal of Finance*, 78(3):1235–1278.

Driessen, J., Lin, T.-C., and Phalippou, L. (2009). A new method to estimate risk and return of non-traded assets from cash flows: The case of private equity funds. NBER Working Paper W14144, National Bureau of Economic Research.

Driessen, J., Lin, T.-C., and Phalippou, L. (2012). A new method to estimate risk and return of nontraded assets from cash flows: the case of private equity. *Journal of Financial and Quantitative Analysis*, 47(3):511–535.

Erel, I., Flanagan, T., and Weisbach, M. S. (2024). Risk-adjusting the returns to private debt funds. Technical report, National Bureau of Economic Research.

Eu, C. N. L. (2017). *Numerical Analysis in Nonlinear Least Squares Methods and Applications*. PhD thesis, Curtin University.

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of financial economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of financial economics*, 128(2):234–252.

Farnsworth, H., Ferson, W., Jackson, D., and Todd, S. (2002). Performance evaluation with stochastic discount factors. *Journal of Business*, 75(3):473–503.

Farrelly, K. and Stevenson, S. (2019). The risk and return of private equity real estate funds. *Global Finance Journal*, 42:100471.

Franzoni, F., Nowak, E., and Phalippou, L. (2012). Private equity performance and liquidity risk. *The Journal of Finance*, 67(6):2341–2373.

Giommetti, N. and Jørgensen, R. (2021). Risk adjustment of private equity cash flows. *Available at SSRN 3980121*.

Gredil, O., Sorensen, M., and Waller, W. (2020). Evaluating private equity performance using stochastic discount factors. Working paper, February 3, 2020.

Gupta, A. and Nieuwerburgh, S. V. (2021). Valuing private equity investments strip by strip. *Journal of Finance*.

Hansen, L. P. and Richard, S. F. (1987). The role of conditioning information in deducing testable restrictions implied by dynamic asset pricing models. *Econometrica*, pages 587–613.

Harris, R. S., Jenkinson, T., and Kaplan, S. N. (2014). Private equity performance: What do we know? *The Journal of Finance*, 69(5):1851–1882.

Hou, K., Mo, H., Xue, C., and Zhang, L. (2019). Which factors? *Review of Finance*, 23(1):1–35.

Hou, K., Mo, H., Xue, C., and Zhang, L. (2021). An augmented q-factor model with expected growth. *Review of Finance*, 25(1):1–41.

Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.

Hüther, N., Schmid, L., and Steri, R. (2022). Credit market equivalents and the valuation of private firms. Unpublished working paper.

Jeffers, J., Lyu, T., and Posenau, K. (2024). The risk and return of impact investing funds. *Journal of Financial Economics*, 161:103928.

Jenish, N. and Prucha, I. R. (2009). Central limit theorems and uniform laws of large numbers for arrays of random fields. *Journal of Econometrics*, 150(1):86–98.

Jenish, N. and Prucha, I. R. (2012). On spatial processes and asymptotic inference under near-epoch dependence. *Journal of Econometrics*, 170(1):178–190.

Jenkinson, T., Landsman, W. R., Rountree, B. R., and Soonawalla, K. (2020). Private equity net asset values and future cash flows. *The Accounting Review*, 95(1):191–210.

Kaplan, S. and Schoar, A. (2005). Private equity performance: Returns, persistence, and capital flows. *Journal of Finance*, 60(4):1791–1823.

Kim, M.-J., Kim, S.-S., and Lee, S.-H. (2015). Estimation of risk and return of venture capital investments in an emerging market: an iterative generalized method of moments approach. *Asia-Pacific Journal of Financial Studies*, 44(3):475–495.

Kim, M. S. and Sun, Y. (2011). Spatial heteroskedasticity and autocorrelation consistent estimation of covariance matrix. *Journal of Econometrics*, 160(2):349–371.

Korteweg, A. (2019). Risk adjustment in private equity returns. *Annual Review of Financial Economics*, 11:131–152.

Korteweg, A. (2023). Risk and return in private equity. In *Handbook of the Economics of Corporate Finance*, volume 1, pages 343–418. Elsevier.

Korteweg, A. and Nagel, S. (2016). Risk-adjusting the returns to venture capital. *Journal of Finance*, 71(3):1437–1470.

Korteweg, A. and Nagel, S. (2024). Risk-adjusted returns of private equity funds: a new approach. *The Review of Financial Studies*, page hhae067.

Korteweg, A., Panageas, S., and Systla, A. (2024). Private equity for pension plans? evaluating private equity performance from an investor's perspective. Technical report, National Bureau of Economic Research.

Kroencke, T. A., Muehler, G., and Sprietsma, M. (2013). Return and risk of human capital contracts. Technical Report 13-108, ZEW-Centre for European Economic Research.

Lee, J. M., Shilling, J. D., and Wurtzebach, C. (2019). A new method to estimate risk and return of commercial real estate assets from cash flows. *Journal of Real Estate Research*, 41(2):183–218.

Li, H., Xu, Y., and Zhang, X. (2016). Hedge funds performance evaluation under the stochastic discount factor framework. *Journal of Financial and Quantitative Analysis*, 51(1):231–257.

Nagar, A. L. (1959). The bias and moment matrix of the general k-class estimators of the parameters in simultaneous equations. *Econometrica: Journal of the Econometric Society*, pages 575–595.

Newey, W. K. and McFadden, D. (1994). Large sample estimation and hypothesis testing. In *Handbook of Econometrics*, volume 4, pages 2111–2245. Elsevier.

Newey, W. K. and Smith, R. J. (2004). Higher order properties of gmm and generalized empirical likelihood estimators. *Econometrica*, 72(1):219–255.

Pötscher, B. M. and Prucha, I. R. (1997). *Dynamic nonlinear econometric models: Asymptotic theory*. Springer Science & Business Media.

Racine, J. (2000). Consistent cross-validatory model-selection for dependent data: hv-block cross-validation. *Journal of Econometrics*, 99:39–61.

Riddiough, T. J. and Wiley, J. A. (2022). Private funds for ordinary people: Fees, flows, and performance. *Journal of Financial and Quantitative Analysis*, 57(8):3252–3280.

Rilstone, P., Srivastava, V. K., and Ullah, A. (1996). The second-order bias and mean squared error of nonlinear estimators. *Journal of Econometrics*, 75(2):369–395.

Tausch, C. and Pietz, M. (2024). Machine learning private equity returns. *The Journal of Finance and Data Science*, 10:100141.

Zhao, J. (2020). Bottom-up portfolio construction of public return factors: Application to european private equity fund benchmarking. Master's thesis, Munich Business School.

	Driessen et al. (2012)	Korteweg and Nagel (2016)	Our approach
Estimator type	Cross-sectional NLS (one-step GMM)	Time-series-like GMM	Nonlinear Least-Mean-Distance
Cash flows priced	PE fund cash flows	PE payoffs + public benchmark moments	PE fund cash flows
Discounting dates	Fund inception only	Fund inception only	Flexible via $\mathcal{T}_i$
What grows	Funds/projects within fixed vintage portfolios	Calendar span with new nonoverlapping payoff windows	Number of vintage years with bounded n_v
What is averaged out	Within-portfolio idiosyncratic fund/project noise	Sampling noise in aggregated SDF-return moments across windows	Short-range dependence across vintages
More data help when	Existing vintages receive more funds	The sample adds new nonoverlapping time windows	The sample adds new vintages, not merely more funds per vintage
Inference	Bootstrap	Spatial HAC	Spatial HAC
Cross-sectional unit	Vintage year portfolio	Single payoff series	Both (tested empirically)
SDF specification	Linear factor model	Exponentially affine	Both (tested empirically)

Table 1: **Comparison of SDF estimation frameworks for private equity.**

training.before	h-block.before	v-block	h-block.after	training.after
estimation	remove	validation	remove	estimation
start-1984	1985,1986,1987	1988,1989,1990	1991,1992,1993	1994-end
start-1987	1988,1989,1990	1991,1992,1993	1994,1995,1996	1997-end
start-1990	1991,1992,1993	1994,1995,1996	1997,1998,1999	2000-end
start-1993	1994,1995,1996	1997,1998,1999	2000,2001,2002	2003-end
start-1996	1997,1998,1999	2000,2001,2002	2003,2004,2005	2006-end
start-1999	2000,2001,2002	2003,2004,2005	2006,2007,2008	2009-end
start-2002	2003,2004,2005	2006,2007,2008	2009,2010,2011	2012-end

Table 2: **Partitions used for hv-block cross-validation.** The number of active folds truncates dynamically based on the maximum vintage year cutoff of the specific estimation run.

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
0 yr	1.242	1.322	0.121	0.94	10.01	1.228	1.163	0.144	1.06	8.09
2.5 yr	1.204	1.152	0.131	1.05	9.00	1.232	1.165	0.174	1.06	6.72
5 yr	1.197	1.146	0.147	1.04	8.01	1.224	1.161	0.199	1.05	5.78
10 yr	1.027	1.618	0.190	0.63	5.43	1.037	1.790	0.234	0.58	4.15
12.5 yr	0.888	1.655	0.227	0.54	4.03	0.929	1.609	0.254	0.58	3.43
15 yr	0.811	1.514	0.227	0.54	3.68	0.857	1.788	0.254	0.48	3.14
17.5 yr	0.749	1.413	0.213	0.53	3.64	0.792	1.690	0.257	0.47	2.86
20 yr	0.696	2.044	0.173	0.34	4.14	0.736	1.738	0.255	0.42	2.66
25 yr	0.681	1.992	0.139	0.34	4.98	0.712	1.682	0.250	0.42	2.59
30 yr	0.681	1.992	0.139	0.34	4.98	0.712	1.682	0.250	0.42	2.59

Table 3: **Empirical estimates for BO (*MKT* factor).** *Notes:* $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = signed stability ratio (estimate / fold dispersion), so negative values reflect negative coefficients and $|SR_{CV}|$ measures the estimate relative to fold dispersion. Significance markers based on t_A are reported for completeness only: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$.

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
0 yr	1.881	1.771	0.393	1.06	4.51	1.588	1.657	0.352	0.96	4.51
2.5 yr	1.966	1.809	0.446	1.09	4.02	1.678	1.657	0.452	1.01	3.69
5 yr	1.998	1.829	0.488	1.09	3.67	1.639	1.655	0.434	0.99	3.68
10 yr	1.645	1.280	0.358	1.28	4.24	1.396	1.948	0.315	0.72	4.39
12.5 yr	1.452	1.279	0.299	1.14	4.49	1.218	1.968	0.150	0.62	7.57
15 yr	1.301	1.356	0.260	0.96	4.55	1.068	6.471	0.138	0.16	6.98
17.5 yr	1.287	1.349	0.275	0.95	4.25	1.046	5.758	0.148	0.18	6.35
20 yr	1.243	1.328	0.278	0.94	4.07	0.986	4.384	0.160	0.22	5.52
25 yr	1.179	1.300	0.267	0.91	4.03	0.902	4.166	0.184	0.22	4.32
30 yr	1.045	1.582	0.262	0.66	3.65	0.817	3.084	0.206	0.26	3.53

Table 4: **Empirical estimates for VC (MKT factor).** *Notes:* $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = signed stability ratio (estimate / fold dispersion), so negative values reflect negative coefficients and $|SR_{CV}|$ measures the estimate relative to fold dispersion. Significance markers based on t_A are reported for completeness only: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$.

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
Panel A: Market Factor (β_{MKT})										
0 yr	1.004	0.845	0.157	1.19	6.04	0.152	28.325	0.698	0.01	0.20
2.5 yr	0.417	1.832	0.404	0.23	1.38	0.277	19.534	0.741	0.01	0.31
5 yr	1.003	2.739	0.333	0.37	2.54	0.398	5.088	0.695	0.08	0.52
10 yr	0.485	1.527	0.351	0.32	1.78	0.437	3.312	0.643	0.13	0.75
12.5 yr	0.348	5.339	0.417	0.07	1.39	0.348	5.102	0.641	0.07	0.69
15 yr	0.354	2.326	0.391	0.15	1.31	0.304	8.993	0.668	0.03	0.60
20 yr	0.454	0.775	0.413	0.59	1.66	0.204^{***}	0.046	0.726	4.41	0.49
Panel B: Alpha ($\hat{\alpha}$)										
0 yr	0.004	0.005	0.002	0.91	1.76	0.008	0.209	0.005	0.04	1.53
2.5 yr	0.006	0.018	0.003	0.35	1.78	0.007	0.141	0.005	0.05	1.28
5 yr	0.003	0.017	0.003	0.16	1.21	0.006	0.040	0.005	0.15	1.16
10 yr	0.004^{**}	0.001	0.003	3.08	1.13	0.004	0.018	0.004	0.24	0.83
12.5 yr	0.004	0.093	0.003	0.05	0.95	0.004	0.036	0.004	0.12	0.78
15 yr	0.004	0.037	0.003	0.10	1.00	0.004	0.074	0.004	0.06	0.70
20 yr	0.002	0.022	0.003	0.09	0.23	0.004^{***}	0.000	0.005	9.79	0.59

Table 5: **Empirical estimates for BO (α + MKT factors).** *Notes:* $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = signed stability ratio (estimate / fold dispersion), so negative values reflect negative coefficients and $|SR_{CV}|$ measures the estimate relative to fold dispersion. Significance markers based on t_A are reported for completeness only: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$.

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
Panel A: Market Factor (β_{MKT})										
0 yr	0.472	18.883	0.188	0.02	3.89	1.741	1.137	0.586	1.53	2.41
2.5 yr	1.426	1.813	0.365	0.79	3.76	1.563	8.719	0.343	0.18	4.62
5 yr	1.476	2.309	0.327	0.64	4.42	1.466	7.777	0.320	0.19	4.83
10 yr	1.801	26.027	0.182	0.07	10.02	1.888***	0.246	0.232	7.69	8.69
12.5 yr	1.977	4.756	0.194	0.42	10.21	2.010	10.488	0.153	0.19	13.64
15 yr	2.136	6.428	0.439	0.33	4.51	2.128	8.550	0.150	0.25	14.49
20 yr	2.293	16.641	0.248	0.14	9.06	2.190	6.838	0.225	0.32	9.82
Panel B: Alpha ($\hat{\alpha}$)										
0 yr	0.020	0.115	0.009	0.17	1.63	-0.001	0.004	0.010	-0.35	0.44
2.5 yr	0.007	0.009	0.005	0.81	1.35	0.001	0.041	0.003	0.03	0.58
5 yr	0.006	0.012	0.004	0.53	1.15	0.002	0.037	0.004	0.05	0.34
10 yr	-0.001	0.123	0.002	-0.01	-1.29	-0.004***	0.001	0.001	-5.13	-4.28
12.5 yr	-0.004	0.007	0.001	-0.65	-5.73	-0.006	0.058	0.001	-0.11	-7.70
15 yr	-0.006	0.036	0.002	-0.18	-2.95	-0.008	0.050	0.001	-0.16	-7.45
20 yr	-0.008	0.088	0.001	-0.09	-9.68	-0.008	0.045	0.002	-0.19	-5.45

Table 6: **Empirical estimates for VC (α + MKT factors).** *Notes:* $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = signed stability ratio (estimate / fold dispersion), so negative values reflect negative coefficients and $|SR_{CV}|$ measures the estimate relative to fold dispersion. Significance markers based on t_A are reported for completeness only: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$.

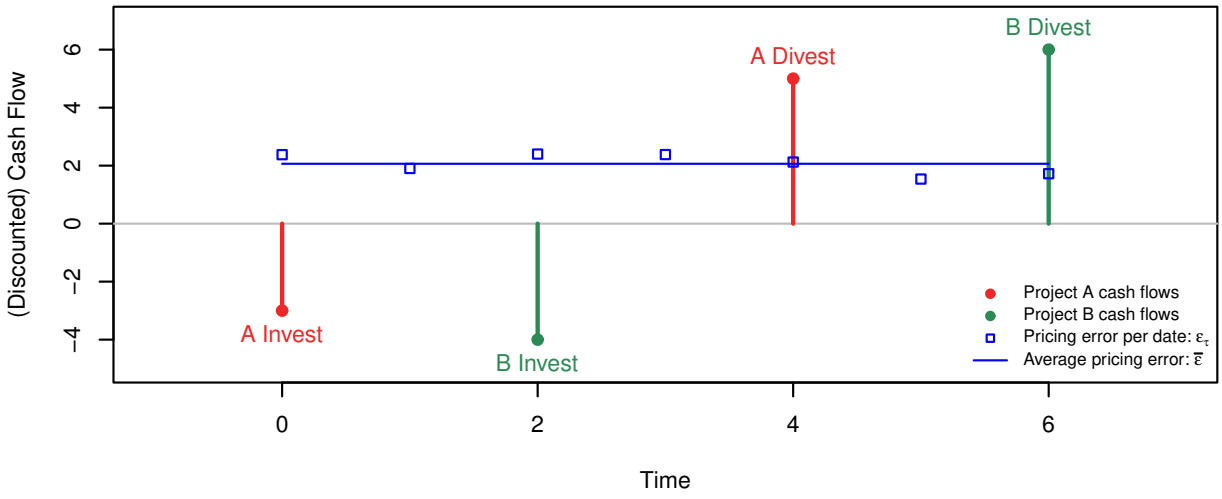

Figure 1: **Interpretation of averaged pricing error.** How to calculate and interpret $\mathcal{T}$-averaging introduced by Equation (11): The time index t is relevant for the cash flows (red and green dots). The time index τ is used for the pricing error, i.e., the sum of discounted net cash flows (blue boxes). The weighted average of these "pricing errors" gives the average pricing error $\bar{\epsilon}$ as defined in Equation (11) (solid blue line). In this example, card($\mathcal{T}_i$) = 7, i.e., the number of blue boxes.

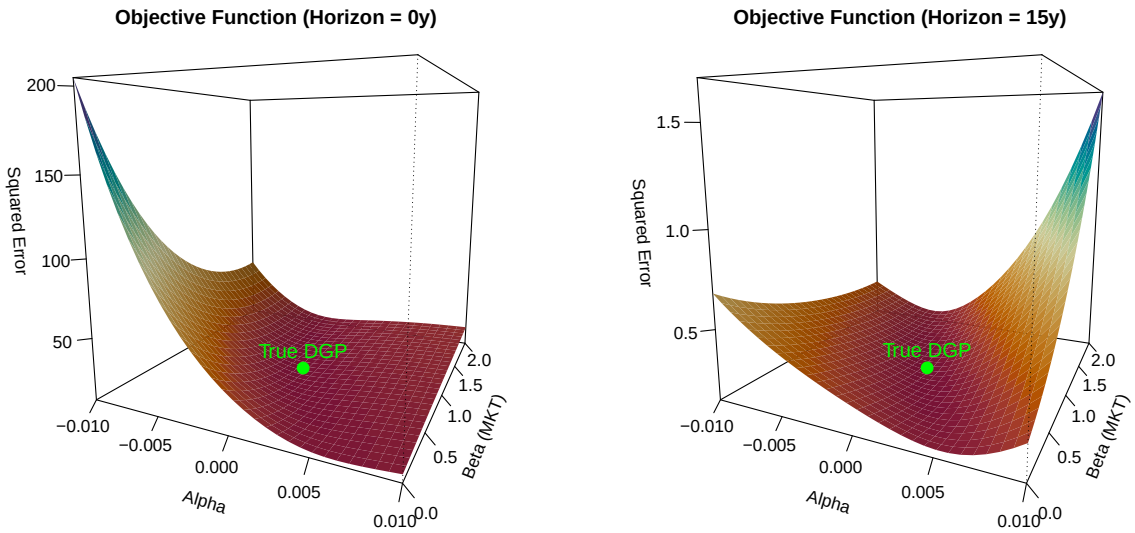

Figure 2: **Regularizing the sample objective function.** The figure plots the sample objective $Q_{n,H}(\theta)$ with quadratic loss over α and β_{MKT}, using simulated cash flows from the base-case DGP (Section 2.5). The green dot marks the true parameters $(\alpha_0, \beta_{\text{MKT},0}) = (0, 1)$. Left panel ($H = 0$): The inception-only criterion suffers from the "exploding alpha" degeneracy. Large positive α values drive multiplicative discount factors to zero, creating a ridge of spurious near-minima along the α axis that prevents reliable identification. Right panel ($H = 15$ years): Horizon averaging regularizes the objective surface, restoring a well-identified minimum near the true parameters.

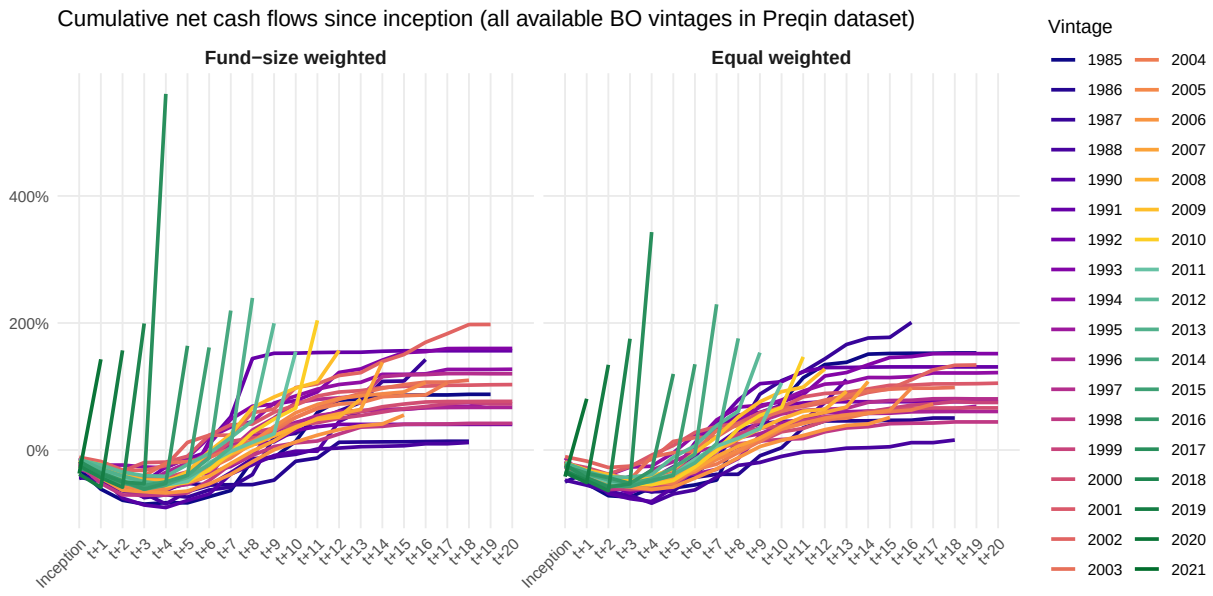

Figure 3: **Cumulative cash flows of BO funds.** J-curves by vintage year for BO funds in Preqin dataset scaled by fund commitment. Post-2010 vintage years are shown as green lines. Based on the sensitivity analysis in Section 2.3, these vintages are excluded from the empirical estimates in Section 3 to limit reliance on unrealized NAVs.

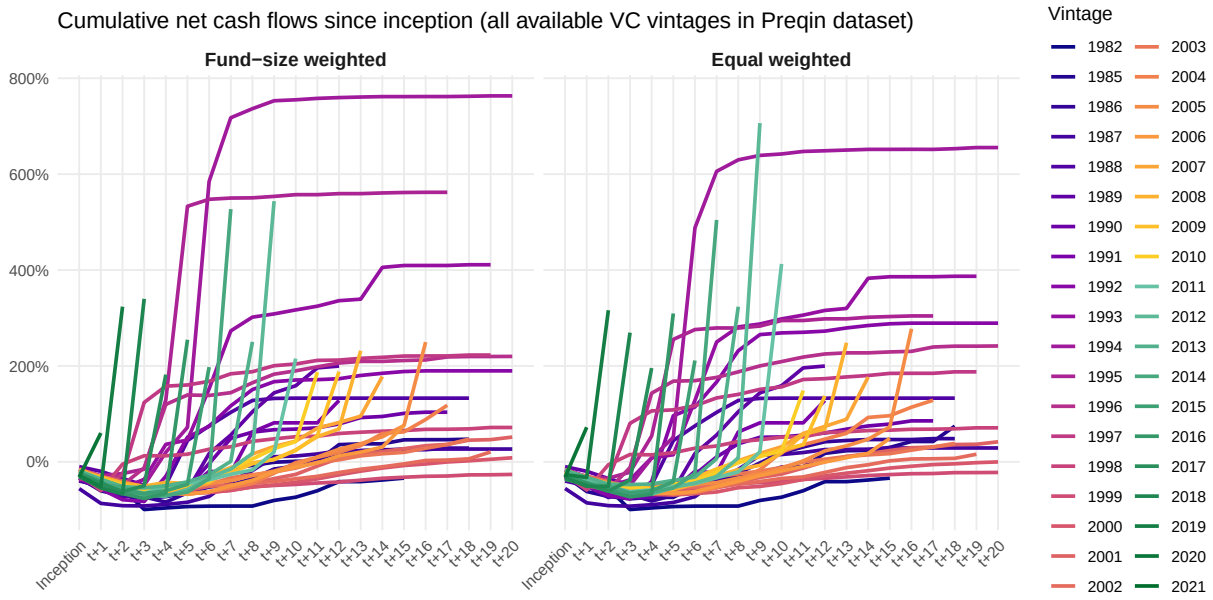

Figure 4: **Cumulative cash flows of VC funds.** J-curves by vintage year for VC funds in Preqin dataset scaled by fund commitment. Post-2010 vintage years are shown as green lines. Based on the sensitivity analysis in Section 2.3, these vintages are excluded from the empirical estimates in Section 3 to limit reliance on unrealized NAVs.

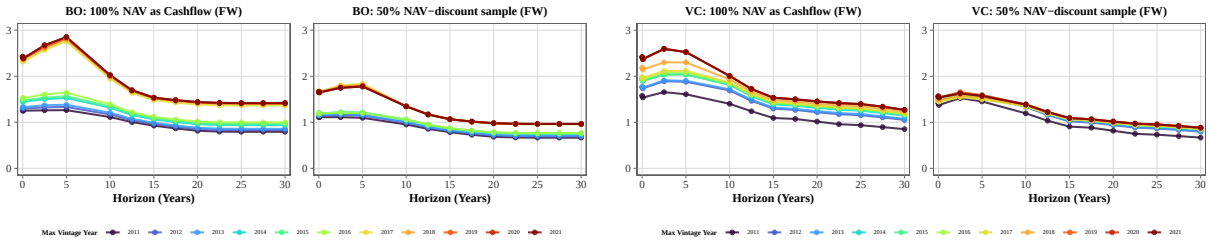

Figure 5: **Sensitivity analysis (MKT factor).** Single-factor model for BO (left) and VC (right) under value weighting, using VYP with different vintage-year cutoffs (asymptotic estimates). For each asset class, the left chart shows the standard baseline (full NAV as final cash flow), while the right chart applies a 50% discount to all final NAVs.

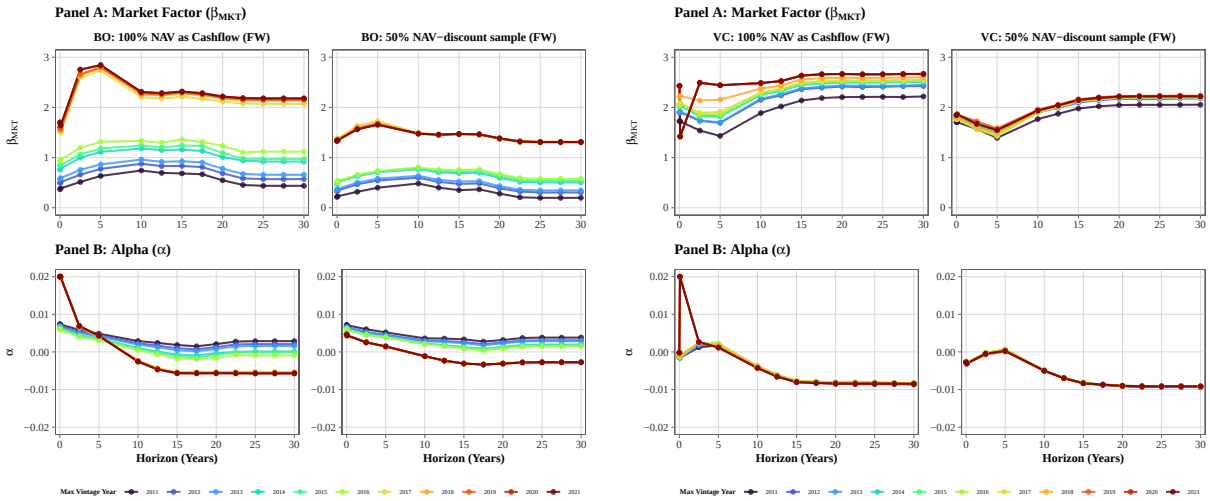

Figure 6: **Sensitivity analysis (α + MKT factors).** Two-factor model for BO (left) and VC (right) under value weighting, using VYP with different vintage-year cutoffs (asymptotic estimates). For each asset class, the left chart shows the standard baseline (full NAV as final cash flow), while the right chart applies a 50% discount to all final NAVs.

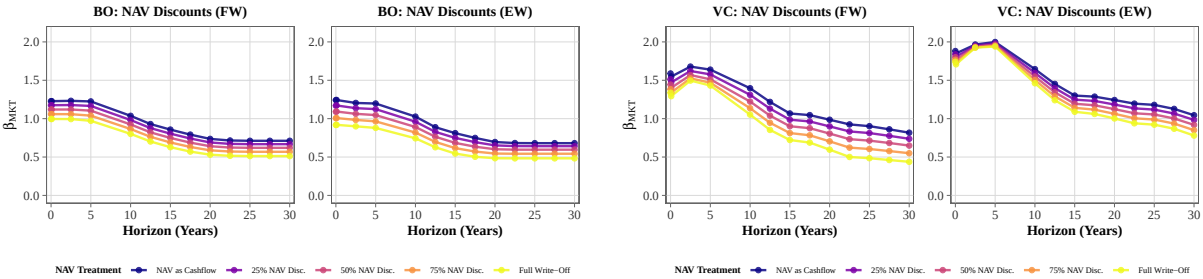

Figure 7: **NAV discount sensitivity (*MKT* factor).** Single-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010. Each panel shows asymptotic estimates under five NAV treatments: full NAV as final cash flow, 25%, 50%, 75% constant discounts, and 100% NAV write-off.

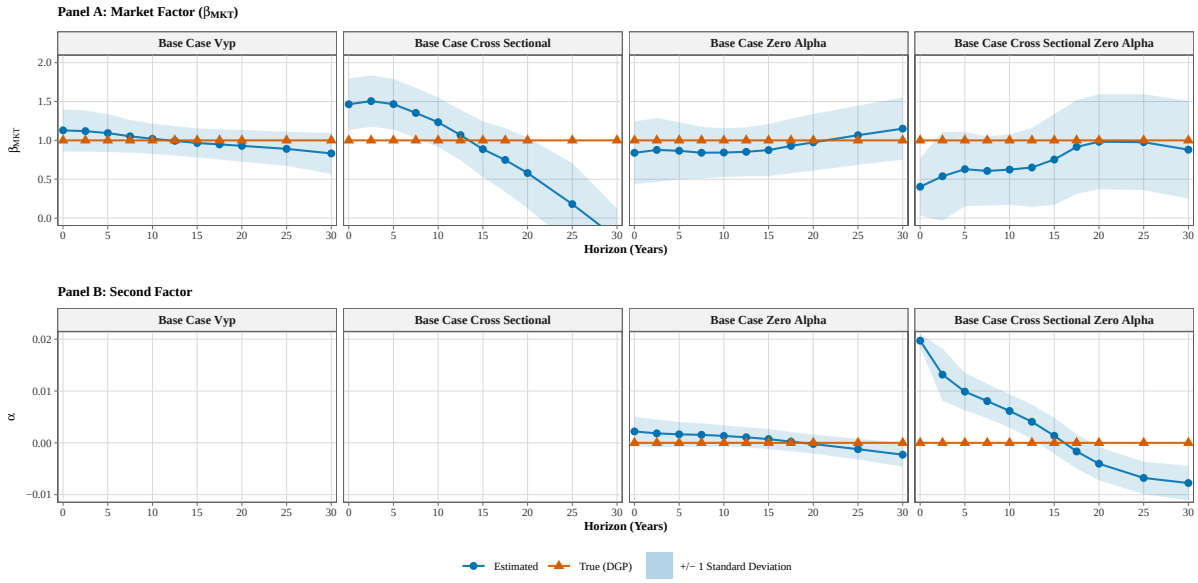

Figure 8: **Simulation results (α + MKT factors).** Single-fund vs. vintage-year portfolios.

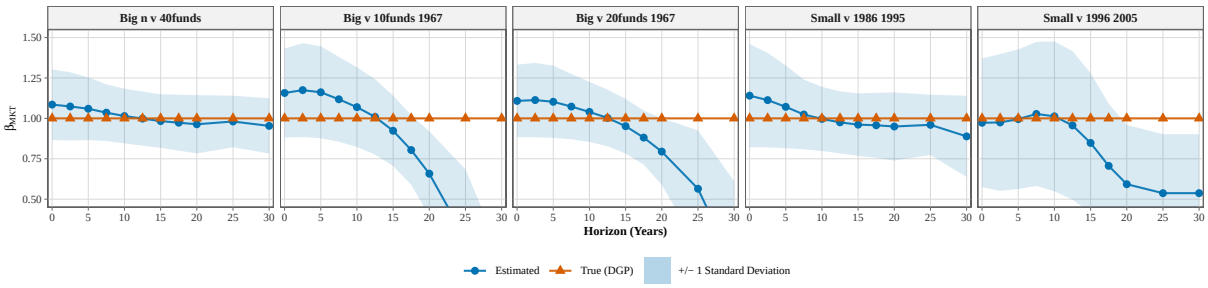

Figure 9: **Simulation results (MKT factor).** Varying vintages V and portfolio sizes n/V.

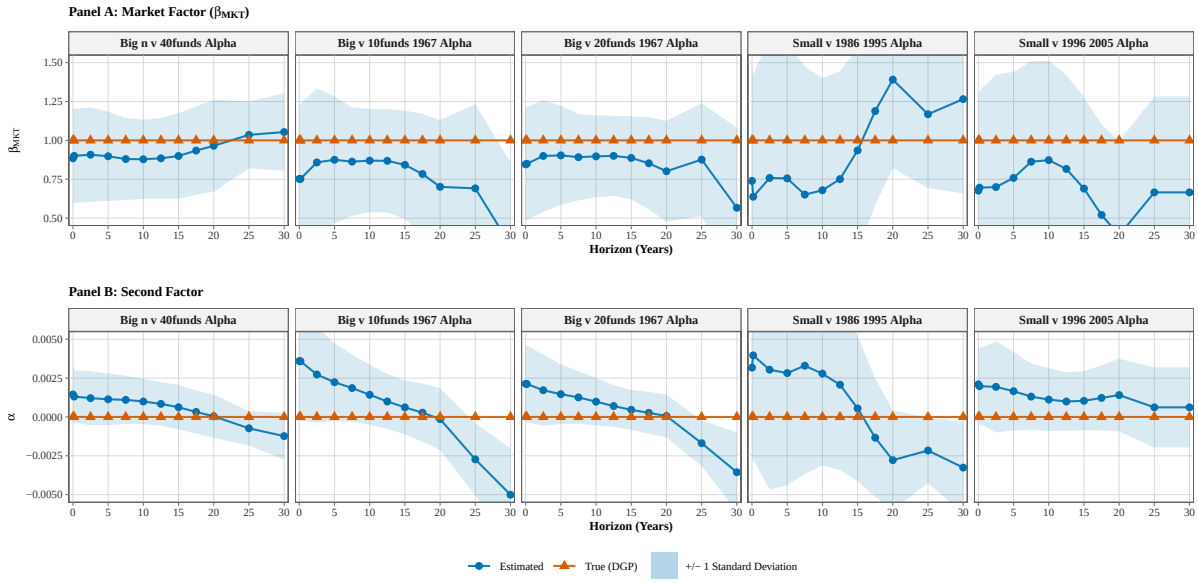

Figure 10: **Simulation results** ($\alpha + MKT$ **factors).** Varying vintages V and portfolio sizes n/V.

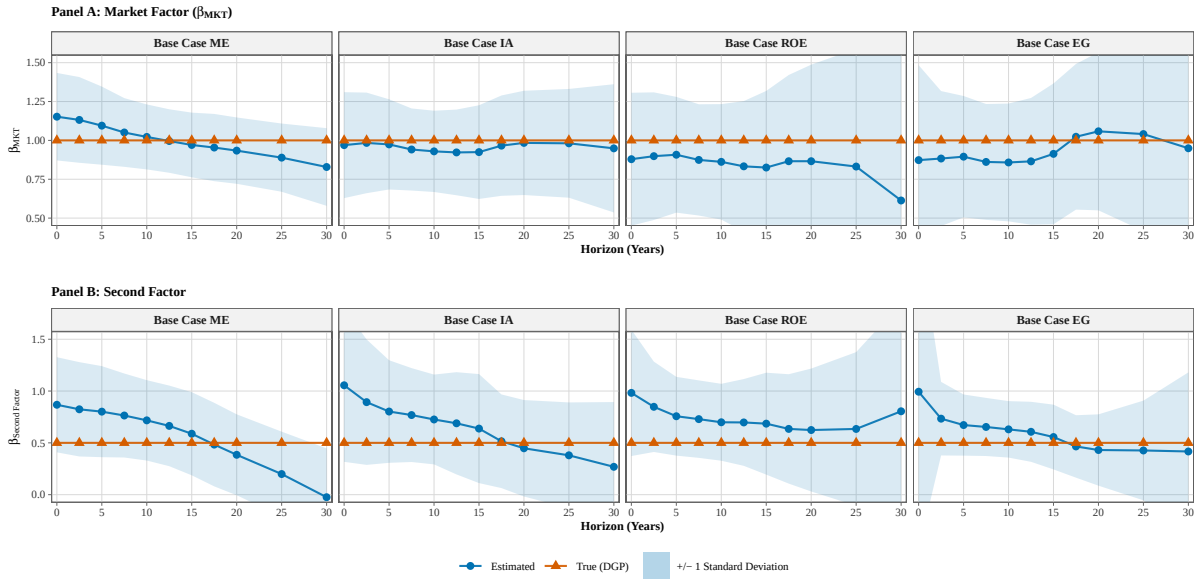

Figure 11: **Simulation results (q^5 factors).** Two-factor models using VYP.

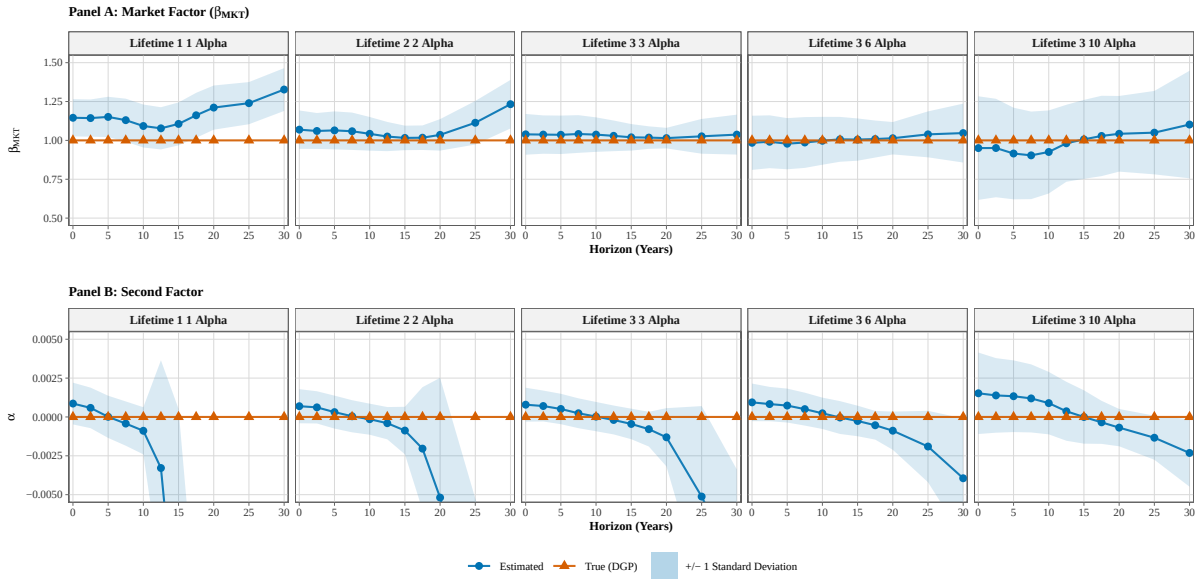

Figure 12: **Simulation results ($\alpha + MKT$ factors).** Varying fund lifetimes using VYP with the following yearly investment-period/holding-period pairs: 1/1, 2/2, 3/3, 3/6, and 3/10 (from left to right).

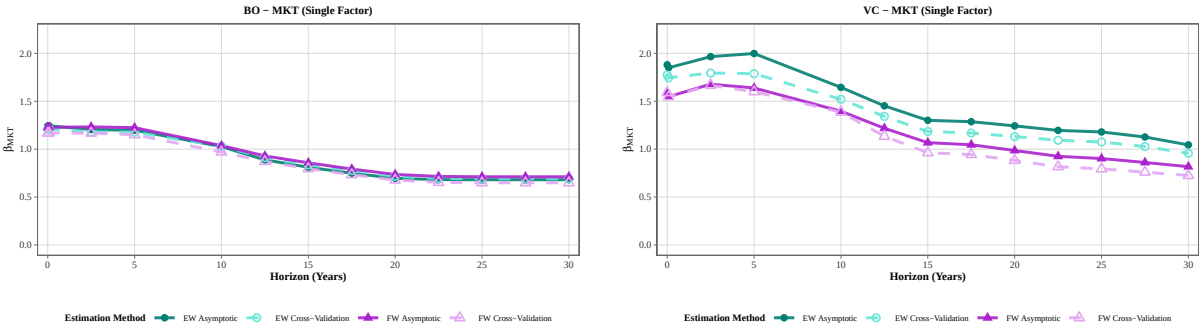

Figure 13: **Empirical results for BO and VC (*MKT* factor).** Single-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010.

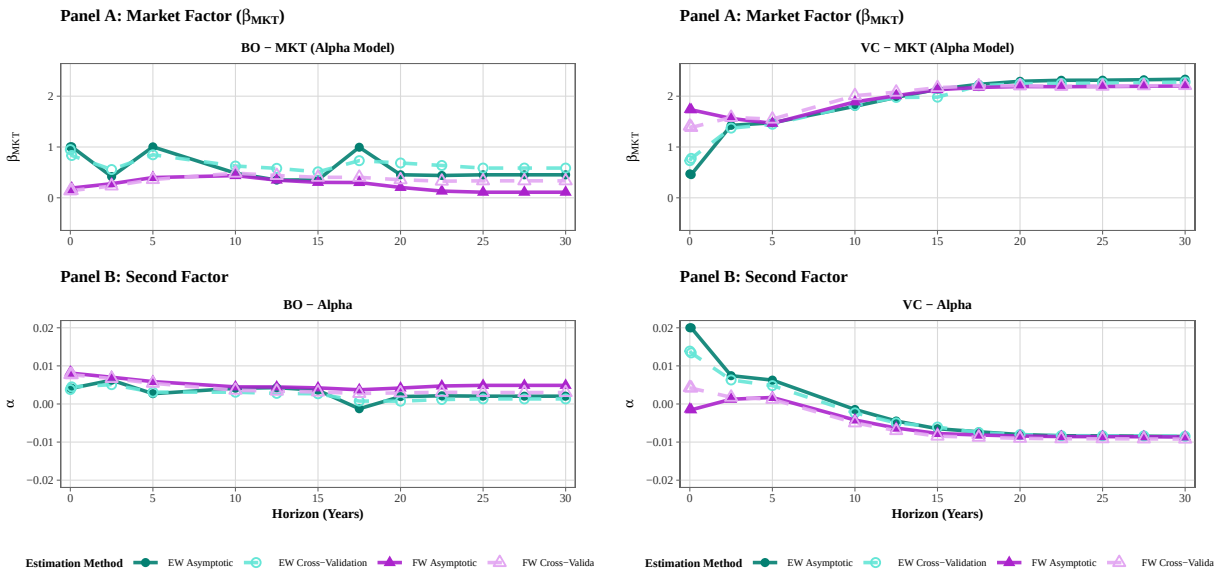

Figure 14: **Empirical results for BO and VC ($\alpha + MKT$ factors).** Two-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010.

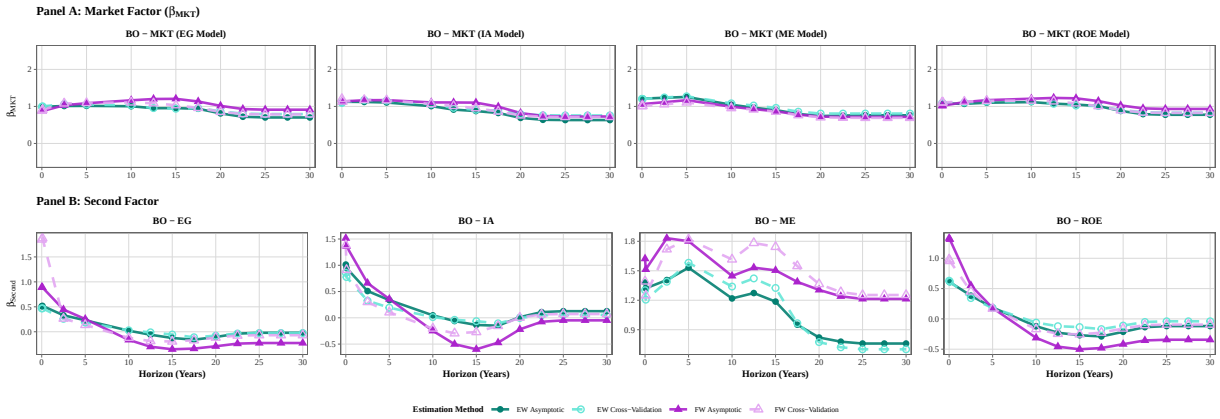

Figure 15: **Empirical results for BO and VC (q^5-factors).** Two-factor model for BO funds using VYP and maximum vintage year 2010.

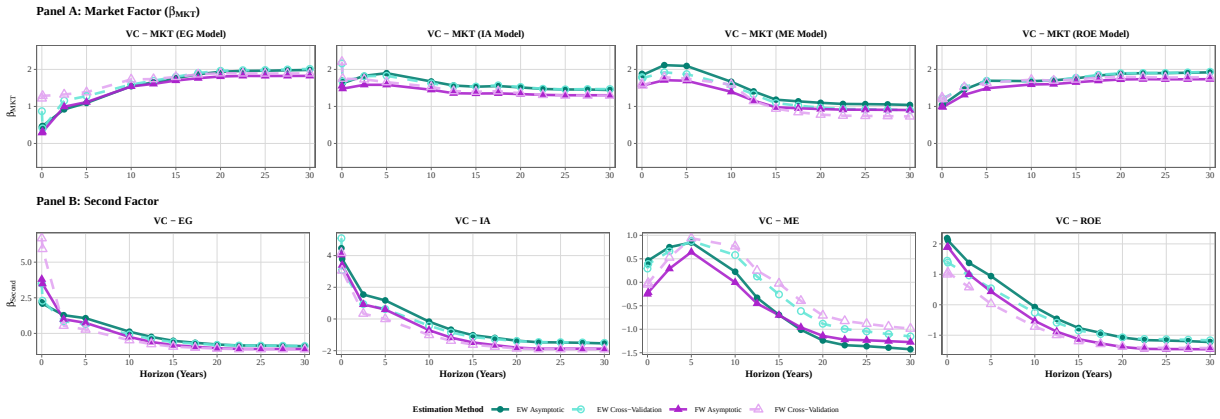

Figure 16: **Empirical results for BO and VC (q^5-factors).** Two-factor model for VC funds using VYP and maximum vintage year 2010.